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Identifying blood vessel boundaries from Flow- MRI data

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Date: 27/5/2026

I hereby declare that, except where specifically indicated, the work submitted herein is my own original work.

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Abstract

This project evaluates three segmentation strategies for extracting blood vessel boundaries from flow MRI scans. These strategies are (i) foundation model with a data prior, (ii) specialist model with a data prior, (iii) model with a geometry prior. For strategy (iii), we propose BETS (Bayesian Estimation of Tubular Segments), a probabilistic model with a geometry prior for inferring the shapes of tubes in the presence of noise. Unlike other geometric models for tube segments, BETS contains a Gaussian process (GP) prior and couples the expectation–maximisation (EM) algorithm with the Laplace approximation for efficient inference and uncertainty quantification. This achieves approximate posterior inference without intractable Markov chain Monte Carlo (MCMC) sampling methods.

We evaluate the three strategies on synthetic and experimental scans. We show that BETS outperforms the foundation model with a data prior on the synthetic scans, and that BETS is the only strategy to produce a plausible prediction on the experimental scan. The specialist model with a data prior performs better than the other two strategies on the synthetic scans when its training distribution matches the testing distribution exactly. Despite this, the specialist model does not generalise realistically to the experimental scan. Of the three strategies, we demonstrate that BETS is the only one to generalise tractably from 3D scans to time-varying 3D scans.

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Notation

Symbol	Description
<i>Scan properties</i>	
I_v	Intensity of the signal within voxel v
$\{I_v\}_{v \in V}$	Full set of scan intensities
$N_x \times N_y \times N_z$	Scan dimensions
g_v	Frangi vesselness filter output at voxel v (Section 2.1)
V	Set of voxels in the scan
$ V $	Number of voxels in the scan
N_v	Number of voxels used for the BETS gradient update steps
h	Length, width and height of the cube-shaped voxels
<i>Parameterisation (BETS)</i>	
$c(s)$	Control line of the tube
$p(s, \theta)$	Location on the tube surface
s	Arc length along the control line
θ	Azimuthal angle around the control line
L	Length of a tube segment
$r(s, \theta)$	Tube radius
$f(s, \theta)$	Log-radius, $r = \exp(f)$
$\kappa(s)$	Curvature of the control line, $\kappa(s) = \ c''(s)\ $
$\hat{e}_r, \hat{e}_\theta, \hat{e}_t$	Radial, circumferential, tangential unit vectors
$\hat{e}_1, \hat{e}_2$	Bishop frame (Section 2.3) vectors perpendicular to $\hat{e}_t$
<i>Harmonic basis and parameters (BETS)</i>	
K, M, T	Truncation orders in s , θ , and (for 4D) t
$a_{km}, b_{km}, c_{km}, d_{km}$	Harmonic coefficients (Equation 4)
$\Phi_j(s, \theta)$	j th basis function (Equation 6)
w_j	j th harmonic coefficient (parameter)
$w_0 = a_{00}$	Constant log-radius mode
w	Vector of harmonic coefficients
N_w	Number of non-constant coefficients, $N_w = (2K+1)(2M+1) - 1$
<i>Gaussian process prior (BETS)</i>	
$k(\cdot, \cdot)$	GP kernel (Equation 7)
σ_f	GP amplitude
ℓ_s, ℓ_θ	Arc-length and angular lengthscales
Λ_{km}	Prior variance of the (k, m) th coefficient
$I_k(z)$	Modified Bessel function of the first kind
Σ_w	Diagonal prior covariance for w
<i>Experimental data likelihoods (BETS)</i>	
σ_n, σ_n^2	Noise standard deviation and variance
$\mu_{\text{in}}, \mu_{\text{out}}$	Inside and outside mean intensities
μ_v	Predicted mean intensity at voxel v

Symbol	Description
p_v	Predicted inside-vessel probability at voxel v
d_v	Radial signed distance for voxel v (Equation 16)
ρ_v	Distance from voxel v to the control line
s_v, θ_v	Coordinates of voxel v projected onto the control line
τ	Sigmoid transition width
$\sigma(\cdot)$	Logistic sigmoid
<i>Multi-vessel and 4D generalisation (BETS)</i>	
$\text{softmin}_\tau\{\cdot\}$	Differentiable approximation to min (Equation 33)
τ_b	Soft-min width for intra-vessel SDF blending
τ'_b	Soft-min width for inter-vessel SDF blending
λ	Strength of the inter-vessel overlap prior
<i>Bayesian inference and EM (BETS)</i>	
$p(\cdot \cdot)$	Conditional probability density
$\mathcal{N}(\mu, \sigma^2)$	Gaussian distribution
ζ	Hyperparameter vector, $[\mu_{\text{in}}, \mu_{\text{out}}, \sigma_n]$
w^*	MAP parameter estimate
$q(w)$	Laplace approximation to the posterior over w
$\mathcal{F}(q, \zeta)$	Evidence lower bound (ELBO)
$\mathbb{E}_q[\cdot]$	Expectation under q
$\mathbb{H}[q]$	Differential entropy of q
H_w	Gauss–Newton Hessian (Equation 22)
J_v	Per-voxel Jacobian row
n	EM iteration index

1 Introduction

1.1 Problem setting

Flow MRI is a medical scanning technique that allows doctors to determine a patient's blood flow in three spatial dimensions and one time dimension. Unlike CT scans, flow MRI does not use ionising radiation. Flow MRI scan times can be prohibitively long because each region must be scanned over many cardiac cycles to suppress noise [1]. Each scanner is also expensive. These two constraints create a bottleneck limiting the number of patients who can be scanned. This limits the income that can be generated.

Recent research has produced an algorithm that takes a noisy scan and infers both the blood vessel boundaries and the blood flow field [2]. The algorithm starts from an initial guess of the blood vessel boundary and applies iterative finite-element method (FEM) and adjoint method updates. Each step produces a candidate boundary and a candidate flow field; the flow field satisfies the Navier–Stokes equations. An optimisation algorithm drives the process towards the most probable boundary and flow field given the noisy scan. This means we can extract an accurate flow reconstruction from brief, noisy scans. This produces a 10–100 times reduction in scanning time [3]. To reduce the number of iterations that the algorithm requires to converge, we need to provide an accurate initial guess of the blood vessel boundary.

This project investigates three strategies to generate this initial blood vessel segmentation. We focus on segmenting aortas and carotid arteries. These are clinically important vessels in the diagnosis and management of cardiovascular disease and strokes. Cardiovascular disease is one of the leading causes of death in the UK [4], and strokes are among the leading costs of care [5]. 3D models of an aorta and a carotid artery are shown in Figure 1, along with a slice of a noisy flow MRI scan of the carotid artery to be segmented.

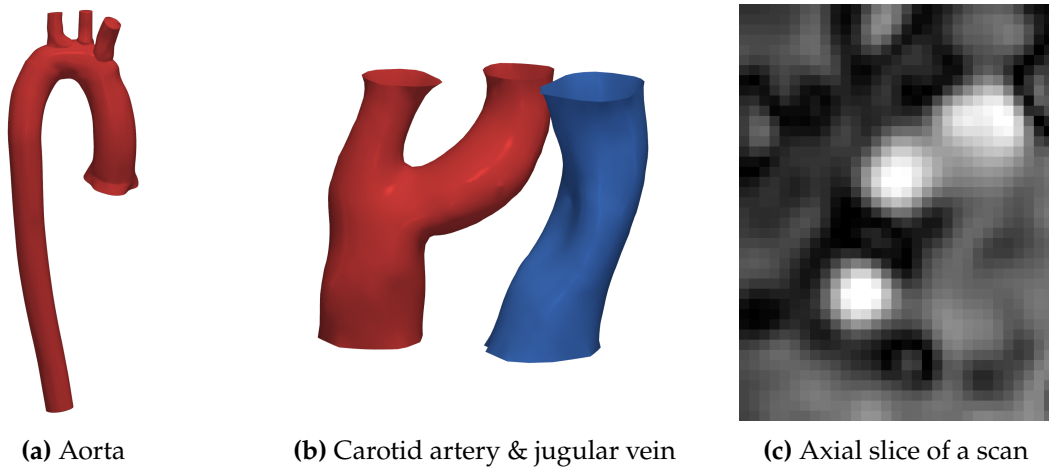


Figure 1: The image in (a) is of a healthy segmented aorta. In (b) the jugular vein, which appears next to the carotid artery, is shown in blue. In (c) the high-magnitude regions (white) correspond to the vessels. Since the scan is of the carotid artery and jugular vein from (b) and there are three high-magnitude regions, the scan is of an axial slice above the branch in the carotid artery.

We aim to use data from short scans, which are low-resolution, such as the scan in Figure 1c. Segmentation is typically performed on data from longer scans that have lower noise levels. Typical MRI segmentation benchmarks focus on developing algorithms to

segment these conventional MRI scans [6]. The high noise levels in Figure 1c mean that visual cues that a traditional segmentation model might typically use to infer the blood vessels, such as the texture of a region, are not present.

1.2 Foundation model with a data prior

A foundation model is a large model that has been trained on an extensive and varied dataset. If necessary, it can then be adapted to downstream tasks by fine-tuning, which is extra training on a specialist dataset [7]. ChatGPT is an example of a foundation model for text generation. It has been trained on a large text corpus and can be fine-tuned for a specific task such as summarising legal contracts. The most notable foundation models for segmentation are the SAM (Segment Anything Model) family from Meta.

Foundation models encode prior knowledge about the problem from their training dataset. We say they have a data prior. Then during inference, they use this prior knowledge to identify the most probable segmentation given the input [8].

We say that models whose prior knowledge about a problem is primarily from explicit geometry constraints have a geometry prior. The model in [2] can be referred to as having a physics prior, because its prior explicitly constrains the output to solutions that satisfy the Navier–Stokes equations.

In this thesis we refer to the alternative to a foundation model as a specialist model. This is a model that has been trained once, on a dataset for a single task in a specific situation. Foundation models for segmentation are increasingly preferred to specialist models for situations where the training data is difficult to annotate. This is because by including segmentation data from many situations (such as segmentations of roads for driverless cars), they can achieve equivalent performance to a specialist model in the desired situation (segmenting flow MRI scans) with fewer training datapoints from the desired situation.

The medical segmentation performance with the SAM family of models is typically moderate before fine-tuning, presumably because medical images are under-represented in the original training dataset [9]. After fine-tuning on medical images, SAM models become state of the art across many medical benchmarks. In this project we focus on MedSAM 2, a version of the SAM 2 model that was adapted to segment 3D medical scans [10].

We did not attempt to fine-tune the MedSAM 2 model any further on specific carotid artery and aorta segmentations because the model is large, and we lacked the resources to train it.

1.3 Specialist model with a data prior

Although fine-tuned foundation models typically match or surpass specialist models at medical segmentation [11], specialist models require significantly less compute to train, making them more accessible [12].

Currently, the U-Net architecture is one of the best-performing specialist models for medical segmentation across many different benchmarks. U-Nets are a type of convolutional neural network with an encoder-to-decoder structure and skip connections

from the encoder to the decoder at matching resolutions [13]. We explain this architecture in more detail in Section 2.2, and show a diagram of it in Figure 3.

We base our research on the nnU-Net scheme. This is an algorithmic way to select the hyperparameters of a U-Net that achieves state-of-the-art performance [12].

1.4 Model with a geometry prior

The flow MRI scans we are segmenting are noisy, which corrupts the information in the scan. This means that prior knowledge is required to filter out the noise. This motivates a switch from models with a data prior to models with a geometry prior so that we can embed as much relevant knowledge as possible into the model.

Since we know that a blood vessel is a tube, we parameterise it by defining a control line that goes through the tube and bifurcates to track the vessel’s branches. We then define the radius to be a function of the position along the control line and the azimuthal rotation around the control line.

We use a truncated harmonic series to describe the variation of the tube’s radius. We then identify the most probable parameters of the harmonic series from the data. We can capture the uncertainty in the segmentation by approximating the probability distribution over the different parameters given the data. We refer to this probability distribution as the posterior distribution. Approximating it means we can identify the regions where the model is least sure about the blood vessel boundary.

Unlike the models with data priors, which are deep learning methods, this is a probabilistic machine learning method. U-Nets and SAM were published recently in 2015 and 2023, but this parameter-fitting segmentation approach was described in 1992 for 2D objects [14]. The computer vision field has generally moved away from the parameter-fitting approach for two main reasons: (i) the vast availability of data means that deep learning approaches are easy to train, and (ii) defining suitable likelihood functions for the parameters to fit is difficult. A likelihood function calculates the probability of the scan given the parameters. A simple likelihood function rejects a lot of useful information, such as image textures, which data-driven methods can leverage [15].

Neither of these objections is applicable to our example. We had very little experimental data available because it takes doctors several hours¹ to segment a flow MRI scan by hand. This made training a deep learning model difficult. Defining a likelihood function was simple because the high noise levels hide most of the informative features of the scans, so a basic likelihood was sufficient.

The BETS (Bayesian Estimation of Tubular Segments) approach that we propose is characterised by:

1. Defining the tube to be the variation of a surface around a control line
2. Parameterising the surface as a harmonic series that approximates a periodic Gaussian process
3. Efficiently optimising the parameters and hyperparameters simultaneously via the EM algorithm
4. Quantifying the prediction’s uncertainty with the Laplace approximation

¹We learnt this through conversations with doctors.

We were unable to find any published research papers that cover all four of these characteristics. The closest match is [16], which uses a similar Gaussian process parameterisation technique. The analysis is only applied to objects with star-shaped boundaries, and characteristics (3) and (4) are replaced by Markov chain Monte Carlo sampling methods that are intractable for tubular boundaries with a high parameter count.

2 Methods

2.1 Foundation model with a data prior

Model selection

For this project we evaluate the performance of MedSAM 2 and SAM 2. MedSAM 2 is a version of SAM 2 that has been adapted to medical segmentation problems. We chose this pair because MedSAM 2 scores higher than the previous state-of-the-art on 3D medical scan benchmarks [10], and evaluating SAM 2 demonstrates the effect of the fine-tuning.

Model architecture

MedSAM 2 extends the SAM 2 architecture with a self-sorting memory-bank mechanism, and its weights are trained on a corpus of medical image-mask pairs [10]. The dataset spans scan modalities such as CT and MRI. The model has five key components, shown in Figure 2 [17]. To make segmentation predictions the model requires a prompt. The prompt we use is to provide lines of points in which each point is guaranteed to be inside a vessel segment. We call these lines the control lines.

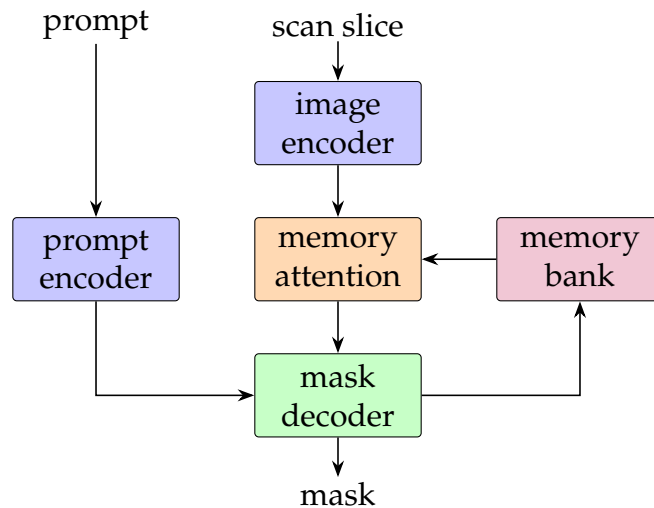


Figure 2: SAM 2 architecture. The image encoder (blue) produces a feature map from each scan slice and the prompt encoder (blue) converts the control-point input into tokens. The memory attention (orange) combines the image features with embeddings cached in the memory bank (purple), and the mask decoder (green) produces a binary mask for the slice. Predictions are written back into the memory bank to inform later slices.

1. **Image encoder.** This is a large vision transformer that ingests each slice of a 3D

medical scan and produces a dense feature map. A vision transformer is a deep learning architecture that breaks the input into small patches of pixels (tokens), and uses an attention layer. The attention mechanism weights the tokens by importance. The feature map is a grid of vectors, where each vector summarises a small image patch.

2. **Prompt encoder.** This is a small deep learning network that converts the control point input into vectors embedded in the same space as the image features.
3. **Memory bank.** This holds spatial memories and prompted-slice memories. A spatial memory is a compressed feature map produced by a small memory encoder that combines the slice’s image features with its predicted mask. A prompted-slice memory is the same as a spatial memory, but only generated from slices where the user supplies a prompt.
4. **Memory attention.** This combines the memories with the information from the current slice to produce memory-conditioned image features.
5. **Mask decoder.** This is a small transformer that maps the memory-conditioned image features plus the prompt tokens to a binary mask for the current slice. The predicted mask and image features are then combined by the memory encoder and fed into the memory bank.

To produce a segmentation of a 3D carotid we identify a control point for each xy slice that we are completely sure is inside each blood vessel. This is straightforward to do by hand, but we automated it by using the Frangi vesselness filter [18] and skeletonising [19]. High outputs from the Frangi vesselness filter mean the pixel is likely to be inside a vessel. Skeletonising means iteratively thinning a binary mask until only a skeleton of control lines remains. The control line extraction is summarised in Algorithm 1.

The Frangi vesselness filter enhances tubular structures by analysing the local geometry encoded in each voxel’s Hessian matrix after smoothing the scan. The Hessian matrix of a bright tube has its smallest-magnitude eigenvalue approximately zero since the intensity has negligible curvature along the vessel axis. The second and third eigenvalues are large and negative, so that the intensity decays quickly perpendicular to the vessel axis.

After generating control points for each slice of the scan we input these as prompts along with the scan’s axial slices to the model. It returns masks that predict the region enclosed by a blood vessel.

- 1: Apply the Frangi vesselness filter to the scan magnitudes $I_v \rightarrow g_v$
- 2: Build a binary mask for vessels by thresholding at the 95th percentile of g
- 3: Retain the two largest connected components to identify the jugular and carotid
- 4: Skeletonise the components

Algorithm 1: Automated control-line extraction from a 3D magnitude scan.

2.2 Specialist model with a data prior

U-Net architecture

Our specialist model is based on the nnU-Net framework, which sets the network’s design from the properties of the training dataset. nnU-Net has won or placed near the top of several MICCAI segmentation challenges [12]. We did not run the original

nnU-Net implementation because it was not supported on our AMD GPU. Although we followed the nnU-Net architecture recommendations, we deviated from the nnU-Net training-loop recommendations to keep the model trainable on the hardware. We also deviated from the nnU-Net data augmentation recommendations because some were unnecessary for our synthetic dataset. We used the MONAI library [20] to implement our U-Net.

The U-Net architecture is a form of convolutional neural network (CNN). This means that it contains convolutional layers. Convolutional layers map an input grid to an output grid by the convolution operation. An example convolution is shown in Equation 1. The values of the kernel are referred to as weights and are chosen to optimise the network’s performance on the training dataset.

$$\underbrace{\begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \end{bmatrix}}_{\text{input } X} * \underbrace{\begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}}_{\text{kernel } W} = \underbrace{\begin{bmatrix} 0 & -4 & -4 & 0 \\ 0 & -4 & -4 & 0 \\ 0 & -4 & -4 & 0 \\ 0 & -4 & -4 & 0 \end{bmatrix}}_{\text{output } Y} \quad (1)$$

Convolutional layers are useful for modelling segmentation because they force translation equivariance and locality [8].

1. Translation equivariance — if a feature is shifted on the input grid it will still provide the same response on the output grid. This is essential for segmentation because if an object is on the left of an image or on the right of an image we want to segment it the same way.
2. Locality — each output grid value is determined by the points near to it on the input grid only. This is essential for segmentation because if an object is on the left of an image only, we do not want pixels on the right of an image to affect its segmentation.

The nnU-Net architecture is composed of convolutional layers. After every two convolution layers we halve the resolution of the activation grid by using a convolution with a stride of length 2. This means that after each dot product operation we move the convolution kernel by two steps, not one. After reducing the resolution to an information bottleneck we then increase the resolution again using layers of transposed convolution. After upsampling, the feature map is concatenated with the skip connection from the corresponding encoder [12]. We display the architecture that we used in Figure 3.

U-Nets are particularly effective at segmentation. The repeated convolutions and downsampling mean that each activation at the information bottleneck is produced from a wide variety of pixels in the input, and produces an abstract, hierarchical representation of the image. This abstract representation is then combined with the input representations on the upsampling side of the network. This contributes pixel-level information. Combining the abstract representation with the pixel-level representation means the segmentation is accurate on a per-pixel basis [8, 13].

We enumerate the architecture, training and data augmentation decisions in Appendix A, and justify where we deviated from the nnU-Net recommendations.

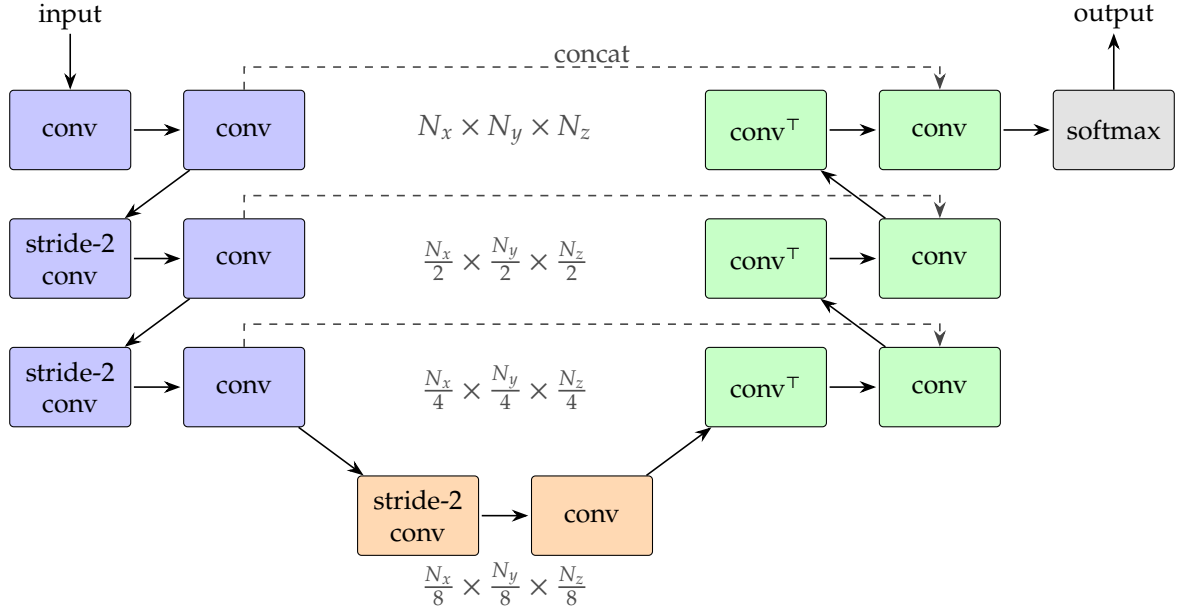


Figure 3: U-Net architecture used in this work. Encoder stages (blue) downsample via stride-2 convolutions to a bottleneck (orange) at one eighth of the input resolution; the decoder (green) upsamples via transposed convolutions (conv^T). Encoder features are concatenated with the matching decoder stage at each resolution.

Training

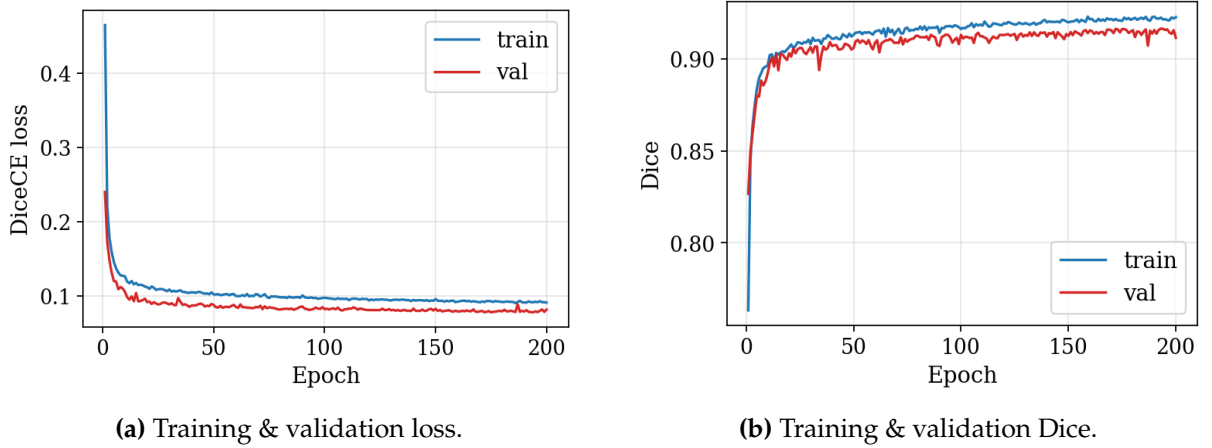


Figure 4: U-Net training curves.

Figure 4 contains the U-Net training and validation curves for the U-Net loss and the Dice score (defined in Section 3.1). The U-Net loss is the cross-entropy loss plus one minus the Dice score. We explain the observations below.

- **Validation loss is lower than the training loss.** Typically the training loss is smaller because the model is fitting the training data. The reason why the training loss is worse for this model is that we train by taking random 32^3 patches and

augmenting them. We then validate using sliding window inference with a Gaussian-weighted overlap. Since validation is an easier task the loss is lower.

- **There is very little overfitting visible.** The training dataset Dice increases only marginally more than the validation dataset Dice over the last 100 epochs. This validates our design choice to use the stochastic synthetic-data generation process as the regulariser (Appendix A, Training table).
- **Most learning happens in the first ~ 50 epochs.**
- **The validation curves are much noisier than the training curves.** This is not a sign of instability, and happens because of the small size of the validation dataset.

2.3 Model with a geometry prior

In this section we describe the geometry-driven model. We describe the concepts below in order.

1. Parameterisation of a single tube segment
2. Prior distribution over that tube segment’s parameters
3. Likelihood of a scan containing the single tube
4. Optimisation to choose the most likely parameters and hyperparameters
5. Uncertainty quantification
6. Generalisation to scans with many tube segments
7. Generalisation to transient 4D flow MRI scans

The BETS strategy is the only strategy we evaluate that generalises to transient flows — the 4D convolutions required for the CNN strategy would be intractable.

Parameterisation

A blood vessel can be split into tube segments. The surface of any tube segment $\mathbf{p}(s, \theta)$ can be described in Equation 2 in terms of a control line that is always within the tube $\mathbf{c}(s)$ and a radius function $r(s, \theta) = \exp[f(s, \theta)]$. The radius function defines the distance between the control line and the tube boundary a distance s along the control line and a rotation θ around it. We define the radius function as the exponential of $f(s, \theta)$ to ensure the radius is always greater than 0.

$$\mathbf{p}(s, \theta) = \mathbf{c}(s) + r(s, \theta) \hat{\mathbf{e}}_r(s, \theta) \quad (2)$$

The unit vectors $\{\hat{\mathbf{e}}_r, \hat{\mathbf{e}}_\theta, \hat{\mathbf{e}}_t\}$ form an orthonormal basis. The vector $\hat{\mathbf{e}}_t$ is parallel to the control line, and $\hat{\mathbf{e}}_r$ and $\hat{\mathbf{e}}_\theta$ are perpendicular to it. This is shown in Figure 5.

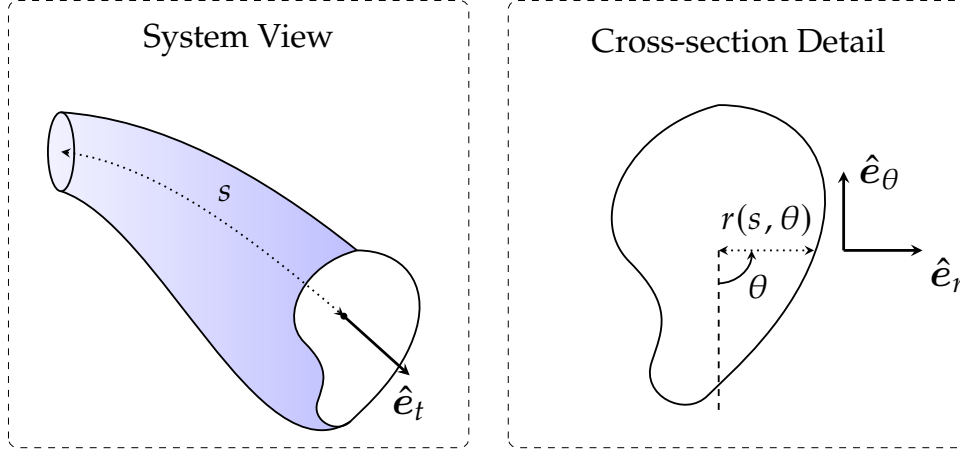


Figure 5: Parameterisation of a single tube segment. Left: the tube with control line, arc-length coordinate s , and tangent vector $\hat{e}_t$. Right: cross-section at fixed s , showing the angular coordinate θ and local basis vectors $\hat{e}_r, \hat{e}_\theta$.

We calculate the orthonormal basis by using the Bishop frame [21] to propagate the unit vectors $\hat{e}_1, \hat{e}_2$ along the control line while minimising their rotation about $\hat{e}_t$. We start by setting $\hat{e}_1(0), \hat{e}_2(0)$ to be two arbitrary orthogonal unit vectors that are perpendicular to the initial tangent. We then iteratively apply incremental updates to calculate $\hat{e}_1(s), \hat{e}_2(s)$ for many different values of s . The Bishop frame update rule means that the vectors smoothly track the control line, remaining perpendicular to it without unnecessary rotation. We then calculate $\hat{e}_r$ and $\hat{e}_\theta$ from Equation 3.

$$\begin{aligned}\hat{e}_r(s, \theta) &= \sin(\theta) \hat{e}_1(s) + \cos(\theta) \hat{e}_2(s), \\ \hat{e}_\theta(s, \theta) &= \cos(\theta) \hat{e}_1(s) - \sin(\theta) \hat{e}_2(s)\end{aligned}\tag{3}$$

We parameterise $f(s, \theta)$ in Equation 4 by a finite linear combination of trigonometric basis functions. Inference of the tube involves identifying the most suitable values of the basis coefficients $\{a_{km}, b_{km}, c_{km}, d_{km}\}$.

$$\begin{aligned}f(s, \theta) = \sum_{k=0}^K \sum_{m=0}^M \bigg\{ \cos\left(\frac{\pi ks}{2L}\right) [a_{km} \cos(m\theta) + b_{km} \sin(m\theta)] \\ + \sin\left(\frac{\pi ks}{2L}\right) [c_{km} \cos(m\theta) + d_{km} \sin(m\theta)] \bigg\}\end{aligned}\tag{4}$$

The constant mode a_{00} is the baseline log-radius. When all the other coefficients are zero, $r(s, \theta) = \exp(a_{00})$, giving a uniform tube of radius $\exp(a_{00})$.

Because $\sin(0) = 0$ we set $b_{k0} \equiv c_{0m} \equiv d_{k0} \equiv d_{0m} \equiv 0$. This gives $(2K + 1)(2M + 1)$ free coefficients. We collect these into a single vector $w = [w_0, w_1, \dots, w_{N_w}]$, where $N_w = (2K + 1)(2M + 1) - 1$. We say $w_0 = a_{00}$ (the constant mode) and $w_1, \dots, w_{N_w}$ are the N_w non-constant coefficients.

To simplify the notation in the next sections, we rewrite the radius function in the compact form in Equation 6. We define Φ_j to be the j th basis function: $\Phi_0 \equiv 1$ is the

constant mode and $\{\Phi_j\}_{j=1}^{N_w}$ are the non-constant trigonometric products. The possible basis functions are enumerated in Equation 5.

$$\begin{aligned} \cos\left(\frac{\pi ks}{2L}\right) \cos(m\theta), & \quad \cos\left(\frac{\pi ks}{2L}\right) \sin(m\theta), \\ \sin\left(\frac{\pi ks}{2L}\right) \cos(m\theta), & \quad \sin\left(\frac{\pi ks}{2L}\right) \sin(m\theta) \end{aligned} \quad (5)$$

$$r(s, \theta) = \exp \left[\sum_{j=0}^{N_w} w_j \Phi_j(s, \theta) \right] \quad (6)$$

To ensure that the cross-section sweep does not fold onto itself, two conditions must be satisfied. Firstly, the vessel surface must always stay below the local radius of curvature: $r(s, \theta) < 1/\kappa(s)$. We define $\kappa(s) = \|\mathbf{c}''(s)\|$. It is the magnitude of the second derivative of the control line $\mathbf{c}(s)$ with respect to arc length s . Secondly, the control line must stay far enough away from itself that distinct parts of the curve do not share closest-point projections. We assume that both of these conditions hold for the scan we are analysing, and do not check or enforce them.

Parameter priors

We know that the tubular vessel has a continuous radius function. It is reasonable to assume that nearby points with similar values of (s, θ) are likely to have similar values of r , and far-away points are less likely to have similar values of r . This means that a Gaussian process with a squared-exponential kernel function would be a good model for $r(s, \theta)$.

A Gaussian process (GP) is a generalisation of a Gaussian random vector to infinite dimensions. It is characterised by a mean function and a covariance kernel. The kernel $k(\cdot, \cdot)$ is the infinite-dimensional equivalent of a covariance matrix. Similarly to how a covariance matrix determines the covariance of two of the terms in a Gaussian vector, a kernel determines the covariance of two points of a Gaussian process. For our modelling we require a kernel function that decreases as the distance between two points increases. This means that points that are close together will have a high covariance and points that are far apart will have a low covariance [22]. We chose to implement this decrease with a squared-exponential function.

By choosing the prior distributions of the coefficients correctly we can use the linear harmonic model to exactly represent a Gaussian process (up to truncation). We choose the Gaussian process with kernel defined in Equation 7. The strength of the decay of the covariance is determined by the hyperparameters ℓ_s and ℓ_θ . The variance is determined by σ_f [23].

$$k((s, \theta), (s', \theta')) = \underbrace{\sigma_f^2 \exp\left(-\frac{4L^2}{\pi^2} \frac{2 \sin^2\left(\frac{\pi(s-s')}{4L}\right)}{\ell_s^2}\right)}_{k_s(s, s')} \underbrace{\exp\left(-\frac{2 \sin^2\left(\frac{\theta-\theta'}{2}\right)}{\ell_\theta^2}\right)}_{k_\theta(\theta, \theta')} \quad (7)$$

The kernel has a periodic structure. This is essential for $k_\theta(\theta, \theta')$ since θ repeats every 2π radians so we force $k_\theta(\theta, \theta') = k_\theta(\theta, \theta' + 2\pi)$. We chose to force a periodic structure on the $k_s(s, s')$ term in the kernel with period $4L$, where L is the length of the tube segment. Forcing a periodic structure means that the Gaussian process has an exact expansion in terms of trigonometric basis functions. Setting the period to $4L$ ensures no unphysical Neumann or Dirichlet boundary conditions are mandated at edges of the domain.

We introduce the $4L^2/\pi^2$ factor in k_s to rescale the periodic kernel so that it matches the standard non-periodic squared-exponential kernel $\exp[-(s - s')^2/(2\ell_s^2)]$ at short distances. This means that ℓ_s keeps its interpretation as a physical lengthscale in voxels, not an angular lengthscale.

To use the linear harmonic model as an approximation of this Gaussian process we set the prior distributions of the basis coefficients to be Gaussian. This is shown in Equation 8, where I_k denotes the standard modified Bessel function of the first kind and δ_{k0} is the Kronecker delta. The constant-mode coefficient $w_0 = a_{00}$ is given a separate prior in Equation 9. This means that the GP amplitude σ_f is not coupled to the baseline radius. We used a mean of $\log(3.0)$ and a variance of 1.0. This gives a 95% range for the baseline radius of $\approx [0.4, 21]$ voxels, which is sufficiently wide for us to use across the evaluation dataset. Because the kernel is separable, the prior is diagonal and the parameters are independent. We set $\sigma_f = 0.3$, which corresponds to $\approx 30\%$ log-radius variation at 1σ ($\exp(\pm 0.3) \in [0.74, 1.35]$).

$$a_{km}, b_{km}, c_{km}, d_{km} [\text{for } (k, m) \neq (0, 0)] \stackrel{\text{indep.}}{\sim} \mathcal{N}(0, \Lambda_{km}),$$

$$\Lambda_{km} = v_k v_m \sigma_f^2 e^{-z_s} I_k(z_s) e^{-z_\theta} I_m(z_\theta), \quad (8)$$

$$z_s = \frac{4L^2}{\pi^2 \ell_s^2}, \quad z_\theta = \frac{1}{\ell_\theta^2}, \quad v_k = 2 - \delta_{k0},$$

$$w_0 = a_{00} \sim \mathcal{N}(\log(3.0), 1.0) \quad (9)$$

Because the standard modified Bessel function of the first kind $I_m(z)$ has the asymptotic in Equation 10, the spectrum decays superexponentially in m . We set $K = 8$ and $M = 4$. The prior variances Λ_{km} have dropped ≈ 1.8 orders of magnitude in s and ≈ 2.4 in θ . This gives $N_w + 1 = 17 \cdot 9 = 153$ modes per segment.

$$I_m(z) \sim \frac{1}{\sqrt{2\pi m}} \left(\frac{ez}{2m} \right)^m \quad \text{as } m \rightarrow \infty, z \text{ fixed} \quad (10)$$

Likelihood and posterior of a scan

We use Bayes' theorem in Equation 11 to identify the most probable parameters given the hyperparameters and the noisy scan intensities $\{I_v\}_{v \in V}$ for each voxel v in the set of voxels in the scan V [24, 25]. We refer to the hyperparameters as ζ and the parameters as w .

$$p(w \mid \{I_v\}_{v \in V}, \zeta) = \frac{p(\{I_v\}_{v \in V} \mid w, \zeta) p(w)}{p(\{I_v\}_{v \in V} \mid \zeta)} \quad (11)$$

$$\propto p(\{I_v\}_{v \in V} \mid w, \zeta) p(w)$$

Bayes' theorem shows that the probability of the parameters given the scan and the hyperparameters $p(w \mid \{I_v\}_{v \in V}, \zeta)$ is proportional to the probability of the scan given the parameters and hyperparameters $p(\{I_v\}_{v \in V} \mid w, \zeta)$ and the prior probability of the parameters $p(w)$. We refer to the probability of the data given the parameters and hyperparameters as the likelihood. To identify the most probable parameters we need to first identify the hyperparameter values, and then maximise $p(w \mid \{I_v\}_{v \in V}, \zeta)$ with respect to w .

The hyperparameters $\zeta = [\mu_{\text{in}}, \mu_{\text{out}}, \sigma_n]$ (these terms are defined later in this section) are unknown. To identify the most probable values of the hyperparameters we maximise the marginal likelihood in Equation 12 over ζ . This means we are choosing the hyperparameters that make the data we observe most likely. This also requires an expression for the likelihood of the scan.

$$\log p(\{I_v\}_{v \in V} \mid \zeta) = \log \int p(\{I_v\}_{v \in V} \mid w, \zeta) p(w) dw \quad (12)$$

We model the probability distribution of a voxel v with intensity I_v in Equation 13. In the BETS algorithm we use a Gaussian distribution, with a mean determined by the radius function, which is itself determined by the parameters and hyperparameters. In reality we expect MRI noise to have a Rician distribution [26]. We use the Gaussian distribution instead because it is similar, and simplifies the model.

$$p(I_v \mid w, \zeta) = \mathcal{N}(I_v; \mu_v(w, \zeta), \sigma_n^2) \quad (13)$$

We are assuming that the noise in the voxel intensities is independent. This is an idealisation since real flow MRI exhibits spatially correlated background noise. We accept the mis-specification because building correlated noise into the model would significantly increase computational cost, and our evaluation shows BETS still outperforms the data prior baselines on the real scan even with the independent noise assumption.

The log probability of an entire scan given the hyperparameters and parameters is shown in Equation 14. We call this the log likelihood. We prefer to use log probabilities because they avoid overflow/underflow errors and make differentiation simpler.

$$\log p(\{I_v\}_{v \in V} \mid w, \zeta) = -\frac{|V|}{2} \log(2\pi\sigma_n^2) - \frac{1}{2\sigma_n^2} \sum_{v \in V} (I_v - \mu_v)^2 \quad (14)$$

The likelihood depends on the parameters w (through μ_v). We define the mean function as a smooth blend between two intensity levels in Equation 15. μ_{out} is the mean value of the background of the scan. μ_{in} is the mean value of the regions of the scan

corresponding to blood vessels. $p_v \in [0, 1]$ represents the probability that the voxel v is inside the blood vessel. For the blood vessel to be identifiable we require $\mu_{\text{in}} \neq \mu_{\text{out}}$.

$$\mu_v = \mu_{\text{out}} + (\mu_{\text{in}} - \mu_{\text{out}}) p_v \quad (15)$$

To model p_v we create a radial signed distance field (SDF) d_v using the function ρ_v , which is the shortest distance from the voxel v to the control line, as in Equation 16. We say the closest point on the control line to the voxel is $c(s_v)$, and has radius $r(s_v, \theta_v)$.

$$d_v = \rho_v - r(s_v, \theta_v) \quad (16)$$

The radial SDF d_v is only an approximation to the actual SDF because it is measured along $\hat{e}_r$ rather than along the true surface normal. The actual SDF at voxel v states the shortest distance from the voxel to a blood vessel boundary.

The radial SDF differs from the Euclidean SDF when the surface normal tilts away from the radial direction. The two distances agree exactly on the surface itself ($\rho = r$), which is where the sigmoid in Equation 17 is most sensitive. The approximation is valid for this project because it only degrades deep inside or outside the boundary. At these points the sigmoid has already saturated so the change to p_v is negligible.

We define p_v using the sigmoid of the SDF, as shown in Equation 17. The term τ captures the length of the transition region at the edges of the tube. We set $\tau = h/4$, where h is the width of a voxel. This means the sigmoid's 10–90% transition width $\approx 4.39\tau \approx h$ matches one voxel.

$$p_v = \sigma(-d_v/\tau) = \frac{1}{1 + \exp(d_v/\tau)} \quad (17)$$

On the one hand, lower values of τ would make the algorithm ill-conditioned. This is because if most voxels saturate then only a thin shell around the border carries the gradient information. On the other hand, larger values of τ would make the boundary smear across multiple voxels. This makes r harder to identify.

In addition to $\zeta = [\mu_{\text{in}}, \mu_{\text{out}}, \sigma_n]$ we have another set of hyperparameters. These are the Gaussian process hyperparameters that describe the Gaussian process prior. Typically we fix the Gaussian process hyperparameters to anatomical values instead of optimising them directly since this reduces the number of iterations required.

Refraining from optimising $\sigma_f, \ell_s, \ell_\theta$ also avoids identifiability issues. The amplitude σ_f and the lengthscales ℓ_s, ℓ_θ trade off in the marginal likelihood. Many $(\sigma_f, \ell_s, \ell_\theta)$ tuples explain the same data variation equally well so the posterior is a ridge rather than a peak. This is a well-known GP identifiability issue [22, 27].

Optimisation

The marginal likelihood. To optimise the scan there are two stages. First we need to choose the hyperparameters that make the observed data most probable. This means maximising $\log p(\{I_v\}_{v \in V} \mid \zeta)$ with respect to ζ . Then we need to choose parameters

that are most probable given the observed data and the hyperparameters. This means maximising $\log p(w \mid \{I_v\}_{v \in V}, \zeta)$ with respect to w .

The integral that defines the marginal likelihood in Equation 12, $\log p(\{I_v\}_{v \in V} \mid \zeta)$ is intractable. The parameters w enter the likelihood through a nonlinear chain $w \rightarrow \log r \rightarrow r \rightarrow d \rightarrow p \rightarrow \mu$, with sigmoid and exponential nonlinearities. We make two approximations to make it tractable.

1. **Laplace approximation.** We replace the true conditional posterior $p(w \mid \{I_v\}_{v \in V}, \zeta)$ with a Gaussian centred at the maximum a posteriori (MAP) estimate.
2. **EM iteration.** We alternate between (i) updating that Gaussian at fixed ζ (the E-step), and (ii) updating ζ at fixed Gaussian (the M-step). The M-step has a closed form. The E-step is a small number of Gauss–Newton iterations to the conditional posterior’s mode. The outer EM loop converges in 5–15 iterations. This is an approximation because the solution that it converges to may not be the global optimum solution.

Both approximations are well known. The Laplace approximation is standard for non-Gaussian posteriors [25, §27]. EM with a Laplace approximation in the E-step is frequently used [24, §4.4 & §9.4], and is referred to as Laplace EM [28].

ELBO. Instead of directly maximising the evidence with respect to ζ , the EM algorithm maximises the ‘Evidence Lower Bound’ which is referred to as the ELBO. We denote the ELBO as $\mathcal{F}(q, \zeta)$. For any proposal Gaussian $q(w) = \mathcal{N}(w; w_q, S_q)$, Equation 18 applies.

$$\begin{aligned} \mathcal{F}(q, \zeta) &= \mathbb{E}_{q(w)}[\log p(\{I_v\}_{v \in V} \mid w, \zeta)] + \mathbb{E}_{q(w)}[\log p(w)] + \mathbb{H}[q(w)] \\ \mathcal{F}(q, \zeta) &\leq \log p(\{I_v\}_{v \in V} \mid \zeta) \end{aligned} \quad (18)$$

The inequality is an equality when $q = p(w \mid \{I_v\}_{v \in V}, \zeta)$. This is the true posterior. For a Gaussian q and a quadratic prior $p(w)$, the only non-trivial term is $\mathbb{E}_{q(w)}[\log p(\{I_v\}_{v \in V} \mid w, \zeta)]$, which we approximate by linearising $\mu_v(w)$ around the centre of q .

EM maximises $\mathcal{F}$ over q and ζ alternately. In the E-step we set $q^{(n+1)} = \arg \max_q \mathcal{F}(q, \zeta^{(n)})$. In standard EM this is the true posterior distribution for w given $\zeta^{(n)}$, but here we use the Laplace approximation. In the M-step we set $\zeta^{(n+1)} = \arg \max_{\zeta} \mathcal{F}(q^{(n+1)}, \zeta)$.

When the E-step is exact, the ELBO is monotonically non-decreasing [29]. With the Laplace E-step this monotonic guarantee is lost. This loss appears not to be problematic in practice because in all the evaluation cases we tested the ELBO did increase monotonically, and convergence was fast (5–15 iterations).

The ELBO $\mathcal{F}(q, \zeta)$ is a lower bound on the log marginal likelihood $\log p(\{I_v\}_{v \in V} \mid \zeta)$, which we refer to as the evidence. The gap between $\mathcal{F}$ and $\log p(\{I_v\}_{v \in V} \mid \zeta)$ is the KL divergence $\text{KL}(q \parallel p(w \mid \{I_v\}_{v \in V}, \zeta))$. The KL divergence is a positive quantity that describes the similarity of two probability distributions. It decreases to zero when q matches the true posterior. This means maximising $\mathcal{F}$ over ζ (when q is a good approximation to the posterior) approximately maximises the scan’s marginal likelihood $p(\{I_v\}_{v \in V} \mid \zeta)$.

E-step. The E-step approximates $p(w \mid \{I_v\}_{v \in V}, \zeta^{(n)})$ as a Gaussian $q(w) = \mathcal{N}(w^*, H_{w^*}^{-1})$. The mode $w^* = \arg \max_w \log p(w \mid \{I_v\}_{v \in V}, \zeta^{(n)})$ is found quickly by Gauss–Newton iteration with the Hessian H_w (Equation 22). H_{w^*} is the Gauss–Newton Hessian evaluated at w^* .

The Jacobian chain $w \rightarrow \log r \rightarrow r \rightarrow d \rightarrow p \rightarrow \mu$ is calculated in Equation 19, and is valid for every $j = 0, 1, \dots, N_w$, including the baseline radius term. For this case, the last factor is 1.

$$\frac{\partial \mu_v}{\partial w_j} = \underbrace{(\mu_{\text{in}} - \mu_{\text{out}})}_{\partial \mu_v / \partial p_v} \cdot \underbrace{[-p_v(1 - p_v)/\tau]}_{\partial p_v / \partial d_v} \cdot \underbrace{(-1)}_{\partial d_v / \partial r_v} \cdot \underbrace{r_v}_{\partial r_v / \partial (\log r_v)} \cdot \underbrace{\Phi_j(s_v, \theta_v)}_{\partial (\log r_v) / \partial w_j} \quad (19)$$

$$\begin{aligned} \mathcal{L}(w) &= -\log p(w \mid \{I_v\}_{v \in V}, \zeta) \\ &= \frac{1}{2\sigma_n^2} \sum_{v \in V} (I_v - \mu_v(w))^2 + \frac{1}{2} (w - \bar{w})^\top \Sigma_w^{-1} (w - \bar{w}) + \text{const}, \\ \bar{w} &= [\log(3.0), 0, \dots, 0] \end{aligned} \quad (20)$$

$$\frac{\partial^2 \mathcal{L}}{\partial w \partial w^\top} = \Sigma_w^{-1} + \frac{1}{\sigma_n^2} \sum_{v \in V} \frac{\partial \mu_v}{\partial w} \frac{\partial \mu_v^\top}{\partial w} - \underbrace{\frac{1}{\sigma_n^2} \sum_{v \in V} (I_v - \mu_v) \frac{\partial^2 \mu_v}{\partial w \partial w^\top}}_{\text{dropped for Gauss–Newton}} \quad (21)$$

The Gauss–Newton approximation of the Hessian of the negative log posterior with respect to w is given in Equation 22, where Σ_w is the diagonal prior covariance. The derivation steps for the true Hessian matrix are shown in Equation 20 and Equation 21. At $j = 0$ the entry is 1.0 for the constant mode prior (Equation 9). It is Λ_{k_j, m_j} for $j \geq 1$ (the GP-derived priors, Equation 8).

$$H_w = \Sigma_w^{-1} + \frac{1}{\sigma_n^2} \sum_{v \in V} \frac{\partial \mu_v}{\partial w} \frac{\partial \mu_v^\top}{\partial w} \quad (22)$$

Two approximations are encoded into our E-step. First we have replaced the true Hessian with the Gauss–Newton approximation of the Hessian. This is exact when the model is linear. For our nonlinear $\mu_v(w)$ the Gauss–Newton matrix and the true Hessian differ by a small residual curvature term.

We expect that at the MAP value the residuals $I_v - \mu_v$ will be zero mean and uncorrelated with the model curvature $\partial^2 \mu_v / \partial w \partial w^\top$, so the dropped term sums to approximately zero. The Gauss–Newton Hessian is therefore an accurate approximation of the true Hessian at the MAP.

The second approximation is the Laplace approximation, which has replaced the true posterior with a Gaussian. Two reasons make the true posterior close to a Gaussian here: (i) the prior on w is exactly Gaussian, (ii) the likelihood is Gaussian in μ_v , which is a smooth function of w , so a quadratic expansion near the MAP is accurate.

M-step for σ_n^2 . At fixed $q(w) = \mathcal{N}(w^*, H_{w^*}^{-1})$ the ELBO as a function of ζ is written in Equation 23.

$$\mathcal{F}(\zeta) = \mathbb{E}_{q(w)}[\log p(\{I_v\}_{v \in V} \mid w, \zeta)] + \text{const} \quad (23)$$

The expectation term expands in Equation 24.

$$\mathbb{E}_{q(w)}[\log p(\{I_v\}_{v \in V} \mid w, \zeta)] = -\frac{|V|}{2} \log(2\pi\sigma_n^2) - \frac{1}{2\sigma_n^2} \sum_{v \in V} \mathbb{E}_{q(w)}[(I_v - \mu_v(w))^2] \quad (24)$$

We compute each per-voxel expectation by linearising the forward model around the MAP, as in Equation 25.

$$\mu_v(w) \approx \mu_v^* + J_v^\top (w - w^*) \quad (25)$$

Substituting, applying linearity of expectation, and using the fact that $q(w) = \mathcal{N}(w^*, H_{w^*}^{-1})$ gives Equation 26.

$$\begin{aligned} \mathbb{E}_{q(w)}[(I_v - \mu_v(w))^2] &= \mathbb{E}_{q(w)}[(I_v - \mu_v^* - J_v^\top (w - w^*))^2] \\ &= (I_v - \mu_v^*)^2 - 2(I_v - \mu_v^*) J_v^\top \mathbb{E}_{q(w)}[w - w^*] + \mathbb{E}_{q(w)}[(J_v^\top (w - w^*))^2] \\ &= (I_v - \mu_v^*)^2 + J_v^\top H_{w^*}^{-1} J_v \end{aligned} \quad (26)$$

To obtain the closed-form σ_n^* that maximises the ELBO, we differentiate it with respect to σ_n^2 . This is shown in Equation 27.

$$\sigma_n^{*2} = \frac{1}{|V|} \sum_{v \in V} [(I_v - \mu_v^*)^2 + J_v^\top H_{w^*}^{-1} J_v] \quad (27)$$

M-step for μ_{in} and μ_{out} . The only term of Equation 18 that depends on μ_{out} and μ_{in} is the likelihood expectation $\mathbb{E}_{q(w)}[\log p(\{I_v\}_{v \in V} \mid w, \zeta)]$, which expands as Equation 28 below.

We approximate the expectation by evaluating the integrand at $w = w^*$, giving $\mathbb{E}_{q(w)}[(I_v - \mu_{\text{out}} - (\mu_{\text{in}} - \mu_{\text{out}})p_v)^2] \approx (I_v - \mu_{\text{out}} - (\mu_{\text{in}} - \mu_{\text{out}})p_v^*)^2$, where $p_v^* = \sigma(-(\rho_v - r(s_v, \theta_v; w^*))/\tau)$ is the occupancy evaluated at the current MAP w^* . This is valid because p_v is a smooth bounded function of w , and we observed empirically by inspecting the Hessian matrix from evaluation cases that the Laplace posterior is sharply peaked at w^* .

We differentiate the resulting objective (Equation 28) with respect to μ_{in} and μ_{out} and set both partial derivatives to zero (Equation 29).

$$\begin{aligned}\mathcal{F}(\mu_{\text{in}}, \mu_{\text{out}}) &= -\frac{1}{2\sigma_n^2} \mathbb{E}_{q(w)} \left[\sum_{v \in V} (I_v - \mu_{\text{out}} - (\mu_{\text{in}} - \mu_{\text{out}}) p_v)^2 \right] + \text{const} \\ &\approx -\frac{1}{2\sigma_n^2} \sum_{v \in V} (I_v - \mu_{\text{out}} - (\mu_{\text{in}} - \mu_{\text{out}}) p_v^*)^2 + \text{const}\end{aligned}\quad (28)$$

$$\begin{aligned}\frac{\partial \mathcal{F}}{\partial \mu_{\text{in}}} &= \frac{1}{\sigma_n^2} \sum_{v \in V} p_v^* (I_v - \mu_{\text{out}} - (\mu_{\text{in}} - \mu_{\text{out}}) p_v^*) = 0, \\ \frac{\partial \mathcal{F}}{\partial \mu_{\text{out}}} &= \frac{1}{\sigma_n^2} \sum_{v \in V} (1 - p_v^*) (I_v - \mu_{\text{out}} - (\mu_{\text{in}} - \mu_{\text{out}}) p_v^*) = 0\end{aligned}\quad (29)$$

This gives the expressions below.

$$\begin{bmatrix} \mu_{\text{in}}^* \\ \mu_{\text{out}}^* \end{bmatrix} = \begin{bmatrix} \sum_v (p_v^*)^2 & \sum_v p_v^* (1 - p_v^*) \\ \sum_v p_v^* (1 - p_v^*) & \sum_v (1 - p_v^*)^2 \end{bmatrix}^{-1} \begin{bmatrix} \sum_v p_v^* I_v \\ \sum_v (1 - p_v^*) I_v \end{bmatrix}\quad (30)$$

Uncertainty quantification

We can rewrite the prediction for $\log r(s, \theta)$ in terms of w in the form of the dot product in Equation 31.

$$\log r(s, \theta) = \phi(s, \theta)^\top w, \quad \phi(s, \theta) = [\Phi_0(s, \theta), \Phi_1(s, \theta), \dots, \Phi_{N_w}(s, \theta)]^\top \quad (31)$$

We approximate the distribution of w given the data and hyperparameters as a Gaussian distribution via the Laplace approximation. Under this approximation, $\log r(s, \theta)$ is Gaussian too. This is shown in Equation 32.

$$\log r(s, \theta) \mid \{I_v\}_{v \in V}, \zeta \sim \mathcal{N}(\phi(s, \theta)^\top w^*, \phi(s, \theta)^\top H_w^{-1} \phi(s, \theta)) \quad (32)$$

This means we can say that under the model there is an approximately 95% chance of $\log r(s, \theta)$ lying within the two-sigma error bounds predicted by this distribution. This enables us to identify which regions of the prediction the model is least certain about.

In reality the errors are greater than the errors predicted by the model. This is because of the mismatch between the model and the scans. The model assumes that there is no correlated noise, which leads it to be overly confident. The uncertainty quantification is still useful, but the error bands must be interpreted as which regions are most certain rather than absolute bounds for the radius.

Generalisation – Multiple tubes

A real scan contains many vessels that cannot overlap. Each vessel consists of one or more tube segments joined at junctions. Each segment has its own arc-length $s \in [0, L_i]$, its own basis truncation, and its own GP coefficients $w^{(i)}$. The radius parameterisation of Equation 4 is unchanged, and is applied independently per segment.

To determine the likelihood we need to compute a single SDF d_v for the whole anatomy. We combine the per-segment SDFs within a vessel using the soft-min function in Equation 33, with smoothing width $\tau_b = h/4$. This is a differentiable approximation to the min function, and blends the segment SDFs smoothly across the junctions within a vessel. We then combine the vessel SDFs into a global scene SDF using the same soft-min function but with a much smaller width $\tau'_b = h/40$, since different vessels should not blend physically.

$$\text{softmin}_{\tau}\{d_i\} = -\tau \log \sum_i \exp(-d_i/\tau) \quad (33)$$

We also introduce an inter-vessel non-overlap prior, given in Equation 34. Without this term, two nearby vessels' occupancies p_A, p_B could overlap, and a voxel could be claimed by both. The prior vanishes when vessels are well-separated and penalises configurations in which two vessels claim the same physical region. The strength λ is set adaptively per scan so that the overlap penalty is roughly ten times the per-voxel likelihood cost of misclassification.

$$\log p_{\text{overlap}} = -\lambda \sum_{A < B} \int p_A(\mathbf{x}) p_B(\mathbf{x}) dV \quad (34)$$

Generalisation – 4D scans

We generalise BETS to 4D by introducing a t variable to the r function, $r(s, \theta, t)$, where $t \in [0, 2\pi)$. We do not vary the control line because although the vessel wall pulsates with the cardiac cycle, the centreline of a vessel typically does not move appreciably.

The kernel of Equation 7 gains a third periodic factor $k_t(t, t')$. This is identical to $k_{\theta}(\theta, \theta')$. The per-segment mode count grows from $(2K + 1)(2M + 1)$ to $(2K + 1)(2M + 1)(2T + 1)$.

3 Evaluation

3.1 Metrics

To evaluate the success of a segmentation we use the Dice score [30], defined in Equation 35. It describes how well set A and set B overlap. It can be expressed in terms of the number of true positives (TP), false positives (FP), and false negatives (FN). True positives are items in both A and B . False positives are items of A not in B . False negatives are items of B not in A . The Dice score ranges from 0 to 1. The segmentation is perfect if the Dice score is 1. In our case the set A is the set of voxels that the model predicts are inside a blood vessel and set B is the blood vessel mask (the set of voxels actually inside the blood vessel).

$$\text{Dice}(A, B) = \frac{2|A \cap B|}{|A| + |B|} = \frac{2 \text{ TP}}{2 \text{ TP} + \text{FP} + \text{FN}} \quad (35)$$

The key problem with the Dice score is that it is insensitive to where the error occurs. To mitigate this weakness we pair it with another metric. We use the 95th percentile

Hausdorff distance, which we refer to as HD95. The symmetric Hausdorff distance is the maximum shortest distance to the opposite surface, over voxels on either surface. This is skewed by the worst case, so we replace the maximum with a 95th percentile (Equation 36). The units of HD95 are voxels [30]. ∂A is the set of voxels on the surface of set A . ∂B is the set of voxels on the surface of set B .

$$\text{HD95}(A, B) = \max \left\{ \text{P}_{95} \min_{a \in \partial A, b \in \partial B} \|a - b\|, \text{P}_{95} \min_{b \in \partial B, a \in \partial A} \|b - a\| \right\} \quad (36)$$

3.2 Performance on synthetic data

We compared the performance of all the models we investigated on a dataset of 10 synthetic scans. We used synthetic scans to quantify performance because we did not have access to any labelled real scans. None of these evaluation scans were used to train the U-Net.

Synthetic data generation

Labelling a scan of the carotid artery is expensive and time-consuming. We learnt from conversations with doctors that it may take several hours. We used Algorithm 2 to generate synthetic scans.

- 1: Build a unit cube grid of size $64 \times 64 \times 64$
- 2: Build the signed distance field (SDF) of a carotid:
 - Create three spline tubes sharing a junction
 - Jitter every control point
 - Sample the trunk radius from a log-uniform distribution
 - Enforce Murray’s law [31] for the branch radii $r_{\text{trunk}}^3 = r_{\text{branch A}}^3 + r_{\text{branch B}}^3$
- 3: Build the jugular vein SDF (a parallel tube along the y -axis with no intersections with the carotid)
- 4: Apply a random morph to every tube by level-set evolution with forward-Euler steps
- 5: Combine the SDFs into the scene SDF by an elementwise minimum
- 6: Create a magnitude image:
 - Define I_v as in Equation 37, where Δ is the half-width of the halo around the SDF zero-level set and C is the inside-vs-outside intensity contrast. Both are randomly chosen for each scan
 - Add Rician noise (the noise distribution of MRI magnitude images) [26]
 - Overlay a Gaussian random field to create large-scale brightness variation
 - Randomly resize the image
 - Crop the image

Algorithm 2: Synthetic 3D carotid scan generation. The algorithm was developed by my co-supervisor Matthew Yoko.

$$I_v = 0.5 C \left(1 - \tanh \left(\frac{3 \text{SDF}_v}{\Delta} \right) \right) \quad (37)$$

Results

Method	Dice mean	Dice range	HD95 mean (vox)	HD95 range (vox)
MedSAM 2	0.709	[0.492, 0.936]	2.16	[1.00, 3.16]
SAM 2	0.696	[0.403, 0.907]	8.22	[1.00, 20.54]
U-Net	0.921	[0.882, 0.957]	1.00	[1.00, 1.00]
BETS	0.804	[0.686, 0.904]	1.93	[1.41, 2.83]

Table 2: Segmentation performance across the 10 synthetic evaluation cases. Range gives the minimum and maximum across cases.

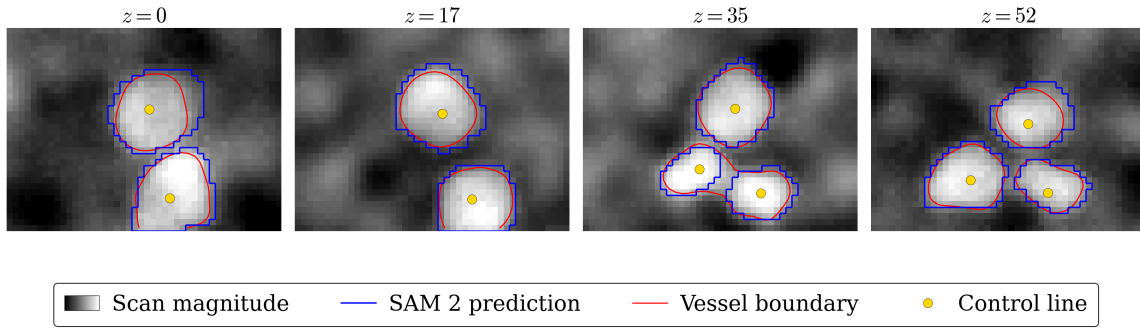


Figure 6: SAM 2, Evaluation case 1. Dice = 0.874, HD95 = 2.00.

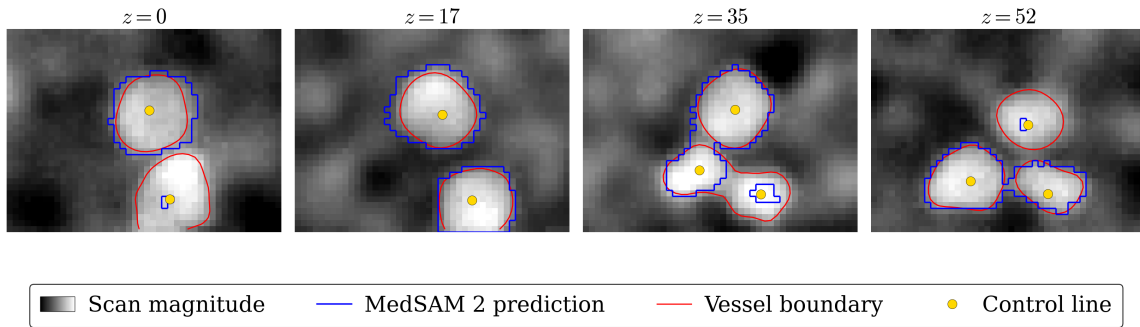


Figure 7: MedSAM 2, Evaluation case 1. Dice = 0.695, HD95 = 3.16.

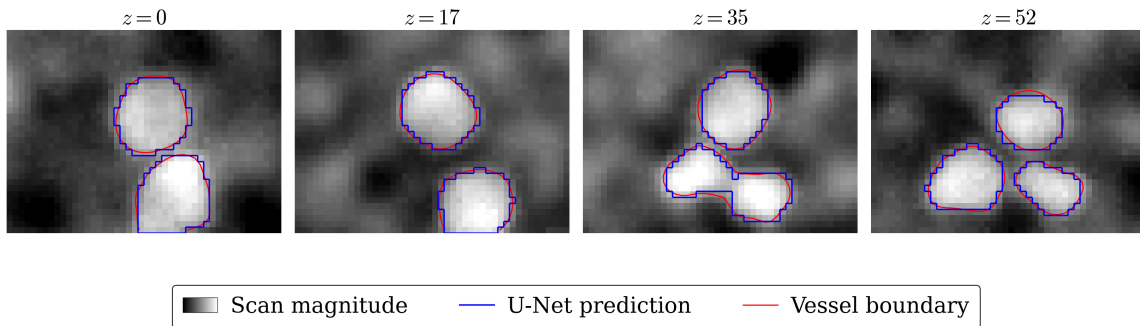


Figure 8: U-Net, Evaluation case 1. Dice = 0.957, HD95 = 1.00.

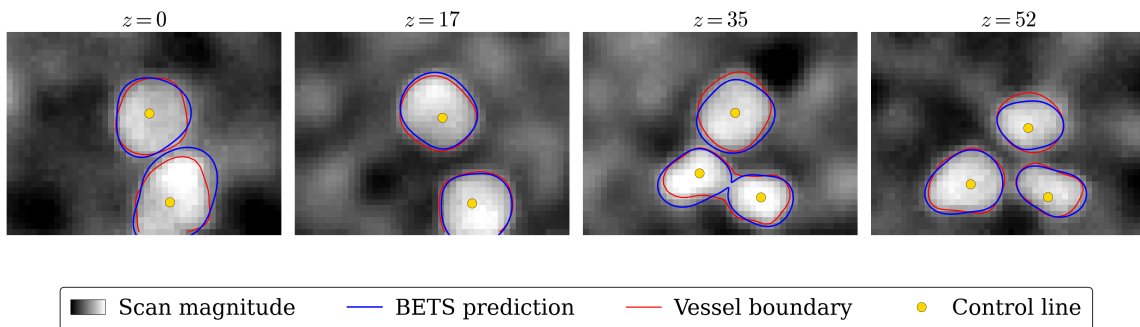


Figure 9: BETS, Evaluation case 1. Dice = 0.904, HD95 = 1.41.

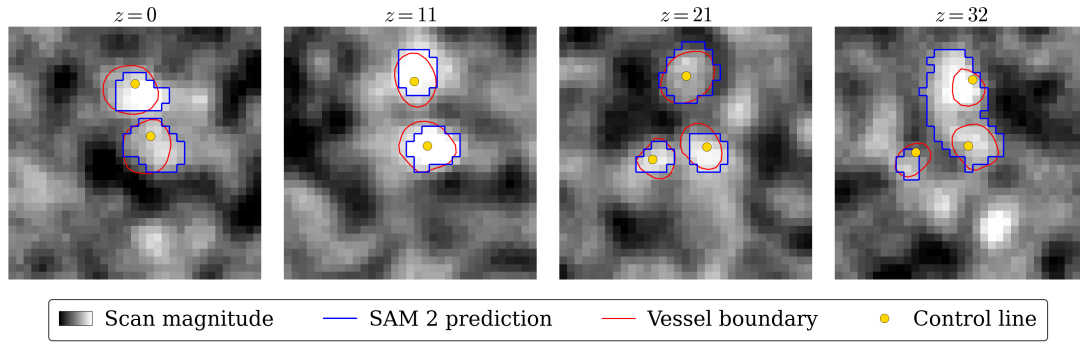


Figure 10: SAM 2, Evaluation case 9. Dice = 0.403, HD95 = 13.30.

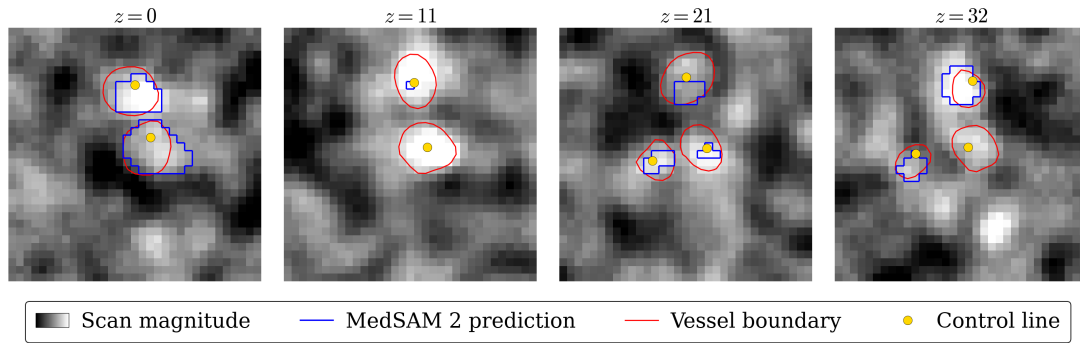


Figure 11: MedSAM 2, Evaluation case 9. Dice = 0.492, HD95 = 2.83.

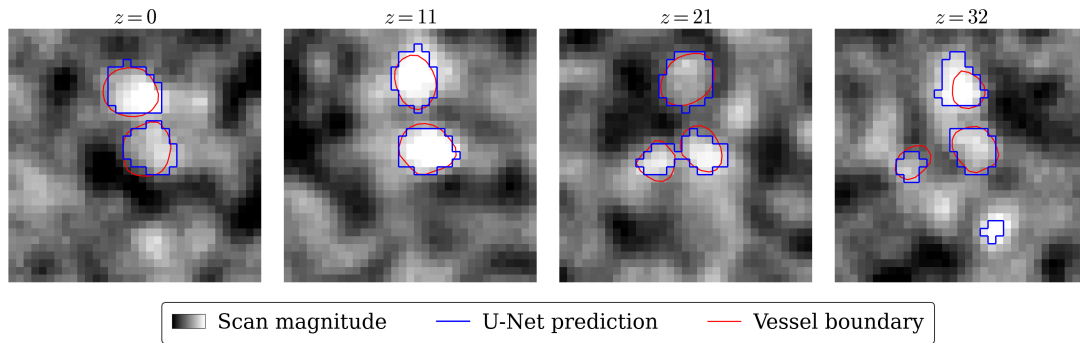


Figure 12: U-Net, Evaluation case 9. Dice = 0.882, HD95 = 1.00.

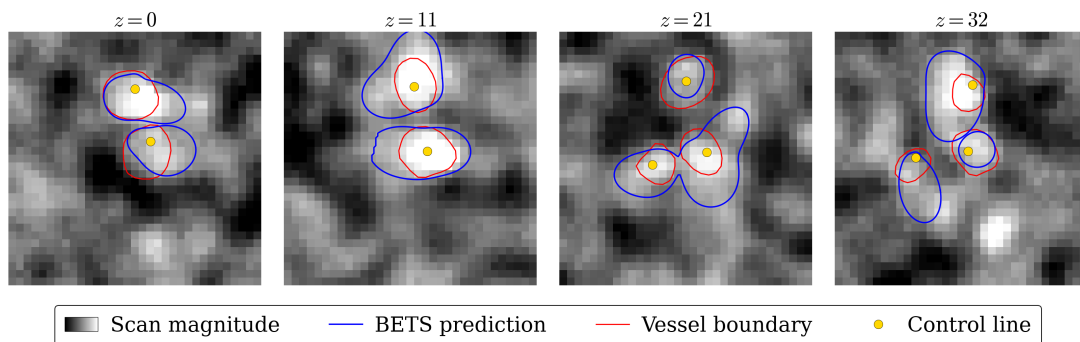


Figure 13: BETS, Evaluation case 9. Dice = 0.686, HD95 = 2.83.

Analysis

On the previous pages we plot slices of the models' predictions from case 1 and case 9 in the evaluation set. Case 1 was the most accurately classified scan, and case 9 was the least accurately classified. Case 1 has a narrow transition region between the blood vessels and the background, and high contrast. Case 9 has a wide transition region between the blood vessels and the background, and low contrast.

Foundation models with data priors (SAM 2, MedSAM 2). MedSAM 2 and SAM 2 are effectively tied on mean Dice score (0.709 vs 0.696). Both foundation models have Dice scores that cover similar ranges. The HD95 score for SAM 2 covers a significantly greater range (1.00 to 20.54) than MedSAM 2 (1.00 to 3.16). This is because SAM 2 has a propensity to include large sections of high-magnitude background material in the scan, as in Figure 10. SAM 2 joins the carotid and the jugular at $z = 32$. This means that we cannot use the segmentation as a starting guess for the algorithm in [2] without separating the segments, which is not straightforward.

MedSAM 2 has a propensity to under-predict the vessel radius. Figure 11 shows that in some cases MedSAM 2 predicts the radius as being a single pixel only. The fine-tuning seems to have suppressed the model's tendency to include background structures in the mask, at the cost of forcing unnaturally small radii.

Although evaluation case 1 is one of the most accurately predicted cases for the other three models, MedSAM 2 fails to achieve a high Dice score on it (0.695). Despite MedSAM 2 and SAM 2 failing in opposite directions, these failures appear to share the same two root causes.

1. **Prior from training data.** MedSAM 2's medically fine-tuned prior appears to have been calibrated to sharp edges. In the ambiguous synthetic scans MedSAM 2's predictions collapse to the high-confidence centre near the control point. Meanwhile, SAM 2's general-object prior latches onto the brightest blobs and over-extends to bright background regions. Neither MedSAM 2 nor SAM 2 has built up a prior distribution from their training data that reflects the type of scans we are segmenting.
2. **3D consistency.** Neither model processes more than one 2D slice at a time. Although the models both store memories about previous slices, they do not have the same level of 3D consistency as a CNN or the BETS algorithm. This makes them fundamentally unsuitable for this task.

Specialist model with a data prior (U-Net). Using synthetic data is not a good way to compare a specialist model architecture with other models because the specialist model will have overfit its parameters to samples drawn from that specific distribution. The training dataset samples were drawn from an identical distribution to the 10 evaluation samples. Here the U-Net acts as an upper bound for what can be inferred from a model with a perfect prior.

The U-Net architecture achieves the best mean Dice (0.921) and best mean HD95 (1.00) on the evaluation dataset. The performance is impressive: the Dice range is tight and high, and the HD95 range is tight and low. Figure 12 shows that for case 9 the U-Net produces a physically unrealistic blob that is falsely segmented as a blood vessel. This

can be identified and removed algorithmically by retaining only the largest connected components in our prediction.

Model with a geometry prior (BETS). BETS achieves a robust second place for classification accuracy and consistency. The lowest Dice score from BETS (0.686) is within a few percent of the mean Dice of MedSAM 2 and SAM 2. Crucially it has a tightly bounded boundary error. The output is always a physically realistic tube with no overlaps between vessels. The predictions’ robustness is due to the strong geometry prior. A tube of varying radius around a fixed control line cannot fragment, and the GP prior prevents unnatural spiky boundaries.

BETS’ key failure mode is radius over-estimation due to high-magnitude background noise that is adjacent to the vessel. This is exacerbated by the transition regions around the blood vessel. Case 9 is an example of this type of failure (Figure 13).

The BETS prediction in Figure 13 would still be an appropriate starting guess for the algorithm in [2] though, since the predicted vessels are in the right regions. In an experimental scan we expect a shorter transition region and greater contrast, so this evaluation case acts as a lower bound for BETS’ real-world performance.

3.3 Performance on real data

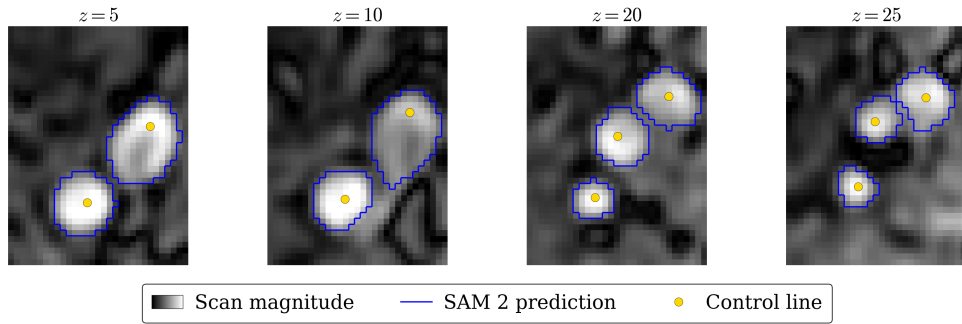


Figure 14: SAM 2, real scan.

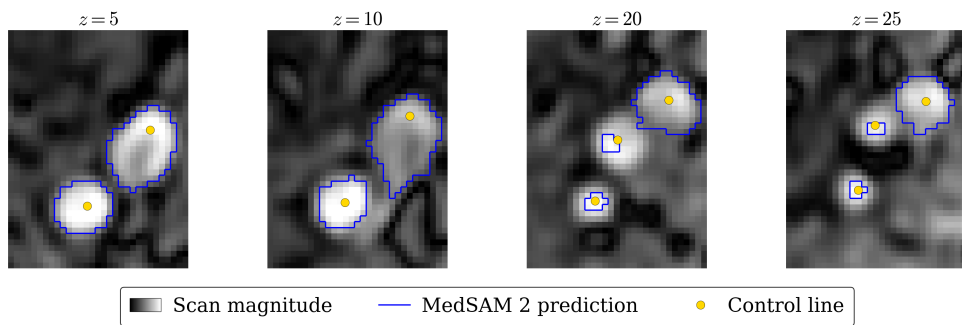


Figure 15: MedSAM 2, real scan.

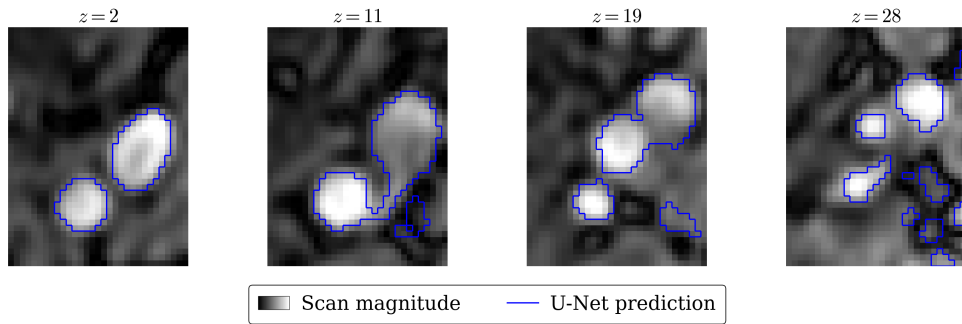


Figure 16: U-Net, real scan.

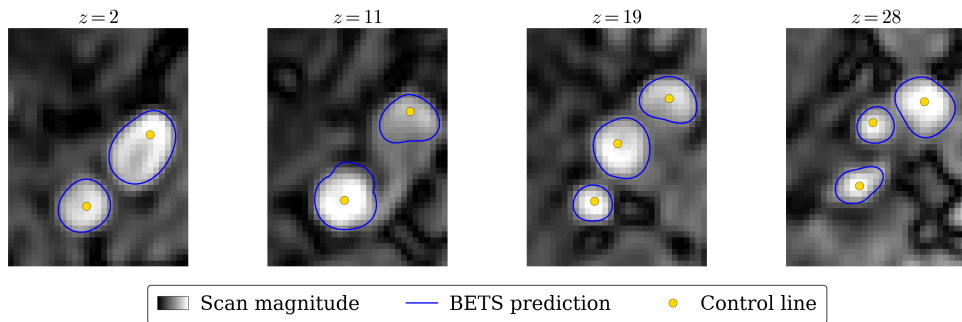


Figure 17: BETS, real scan.

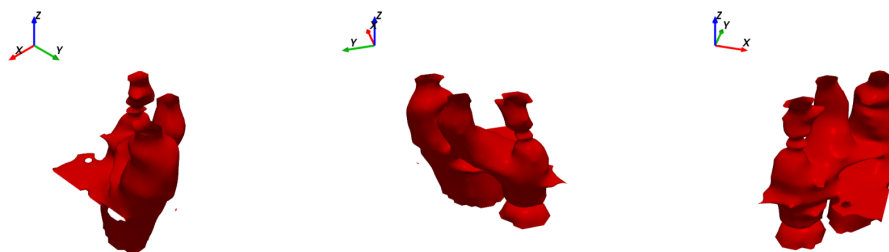


Figure 18: SAM 2, real scan 3D reconstruction.

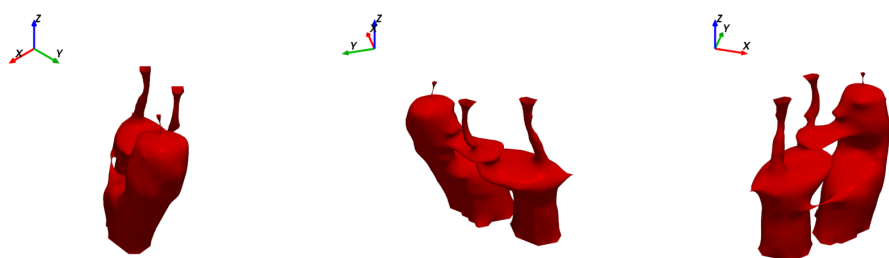


Figure 19: MedSAM 2, real scan 3D reconstruction.

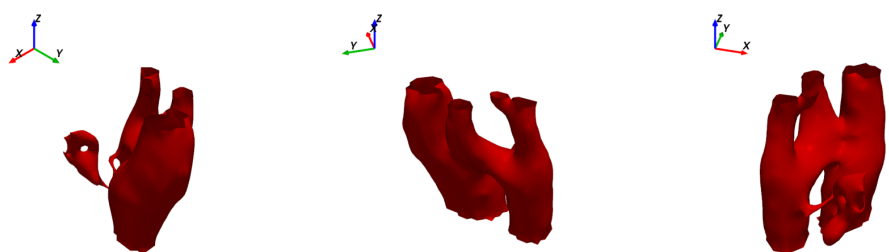


Figure 20: U-Net, real scan 3D reconstruction.

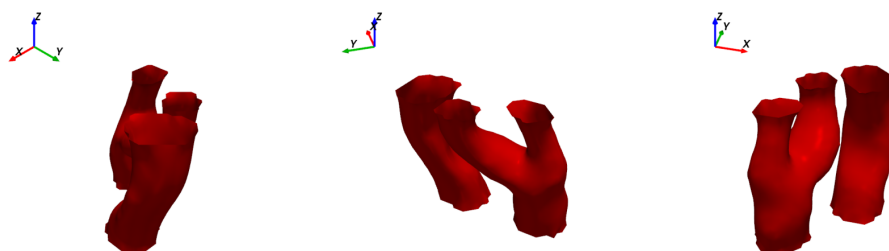


Figure 21: BETS, real scan 3D reconstruction.

Since we did not have access to any labelled scans, we have not quantified the segmentation accuracy on the real scan we tested on. The results of this section are more relevant than those of the previous section because in practice the data we segment will be experimental data, not synthetic data.

Foundation models with data priors (SAM 2, MedSAM 2). Both the foundation models fail catastrophically (Figures 18 and 19). The tube widths expand to fill significantly more of the scan than the blood vessels ought to, and the radius varies drastically between slices (Figure 14). This poor performance has the same two causes as on the synthetic cases. The fact that MedSAM 2 over- and under-predicts nearby slices of the same vessel in Figure 19 is direct evidence that the slice-by-slice architecture lacks sufficient 3D consistency.

Specialist model with a data prior (U-Net). Although the outlines of the vessels shown in Figure 16 seem sensible at first glance, much of the background tissue is segmented into blobs that are mistaken for blood vessels. In Figure 20 we have removed all of these blobs by retaining the largest connected component only.

The U-Net has connected the jugular and carotid blood vessels. Figure 16 shows paths between them at $z = 11$ and $z = 19$. This makes its outputs an unacceptable starting guess for the algorithm in [2]. This drop in performance versus the evaluation cases is because the real scan is not drawn from the same distribution as the training data². Another problem with the segmentation is that it introduces a non-existent branch at the top of one of the carotid branches.

Model with a geometry prior (BETS). The BETS reconstruction is the cleanest (Figure 21). BETS was uncertain about the predictions for radii at the top and bottom of the scan, so we have cropped the top and bottom 3 slices of voxels. The lack of certainty is partly because inference is harder at the borders. The model has half as much information about the local behaviour of the tube as it does in the middle of the scan, since the GP only has neighbours in one direction along s .

An extra caveat with BETS is the scalability of the algorithm. The complexity of the E-step is $O(N_v N_w^2)$. N_w is the number of parameters, and N_v is the number of voxels that we include in the gradient computations, $N_v \gg N_w$. On the $64 \times 64 \times 64$ scans a fit takes 30 to 90 seconds. This is not a significant barrier to implementing the algorithm on larger scans, though. If necessary, a large scan of a blood vessel could be broken into a sequence of smaller scans to be segmented, and the results could be stitched together.

²Surprisingly, we can prevent the U-Net from joining the jugular and carotid by increasing the contrast and the vessel transition lengths of the scans in the synthetic dataset far past the physical levels. This means that the U-Net will perform worse on the evaluation set, but its real-world predictions will be more plausible. Our theory is that including more noisy and uncertain cases in the training dataset means the U-Net is less confident in its predictions, and learns to trust its prior more and the data it is shown at inference time less. This means that it is not influenced so much by the out-of-distribution structures in the experimental scan. We did not include this U-Net in the report because its predictions were still too non-physical to use as a starting guess for the algorithm in [2] so our conclusion does not change. Although the experimental prediction was more realistic and matched the BETS prediction more closely, it involved a hole through the jugular. This model would not count as a specialist model because we are training the model from cases that are significantly different to the testing cases.

We do not include most of the voxels in the scan in the gradient calculations because most are at saturated sigmoid values and do not contribute any gradient information. This means that if the scan size is doubled to include twice the length of tube, we expect N_v , and hence the run time of the algorithm, to only increase by a factor of 2.

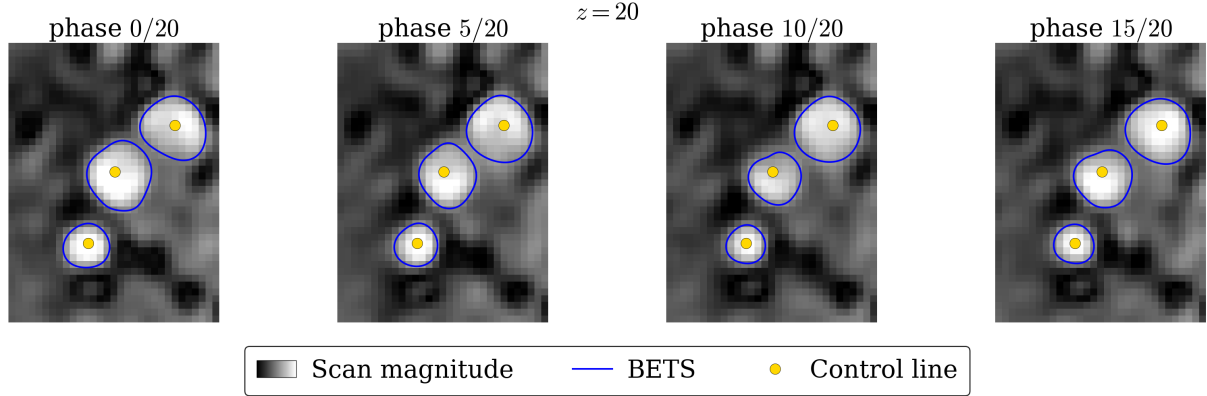


Figure 22: BETS, real 4D scan, four time-frames at $z = 20$.

Figure 22 shows the BETS model applied to a transient scan. We show 4 of the 20 frames that cover one cardiac cycle. Each time-frame of the image is shown at $z = 20$. The control lines are fixed across the cycle, but the surface of the blood vessels deforms over time.

The fits stay consistent across the cycle, with no collapse and no large jumps. Because the model forces smooth variation between frames, we expect that the 4D segmentations are more accurate than the 3D ones, since we have more information and can ignore more of the noise.

3.4 Uncertainty quantification for the geometry prior model

We mentioned in the previous section that we had cropped the top and bottom of the experimental scan. Figure 23 shows the uncertainty plot of the jugular vein that led us to do this. In addition to having less information at the edge of a scan, some of the radius predictions at the top and the bottom correspond to voxels outside the scan, since the control lines are not parallel with the z -axis at the boundaries. For these sections the model is using its smoothness prior only, since there is no data to condition the predictions on.

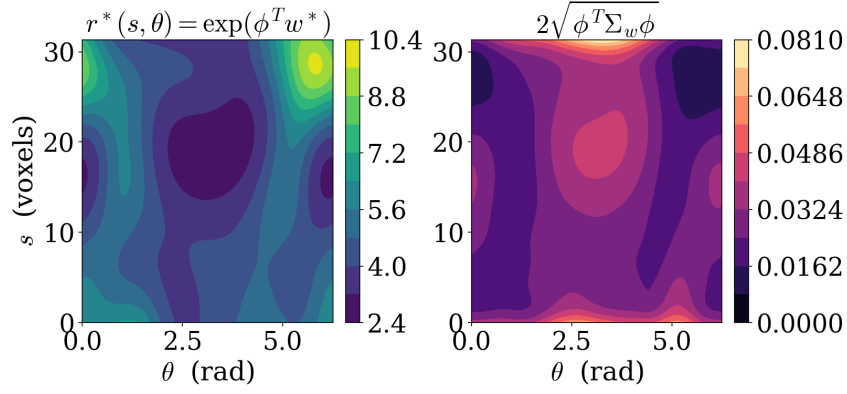


Figure 23: Uncertainty quantification for the jugular vein of the real scan, plotted over arc-length s (voxels) along the control line and azimuthal angle θ around the tube. Left: the MAP radius $r^*(s, \theta)$. Right: the two-standard-deviation predictive band on the log-radius. The real scan has no ground-truth mask, so no error panel is shown.

Figure 24 shows the uncertainty for the jugular vein of the first evaluation case. The high-uncertainty regions do not align with the high-error regions, because of model mismatch. BETS assumes in its likelihood function that the noise is not correlated, which is a false assumption. When it encounters correlated background noise near a blood vessel it treats it as far less probable than it ought to, skewing its confidence.

For the edges of the domain where $s = 0$ and $s = L$ it is impossible to calculate the error. These are high-uncertainty regions for this scan, as they were for the experimental scan.

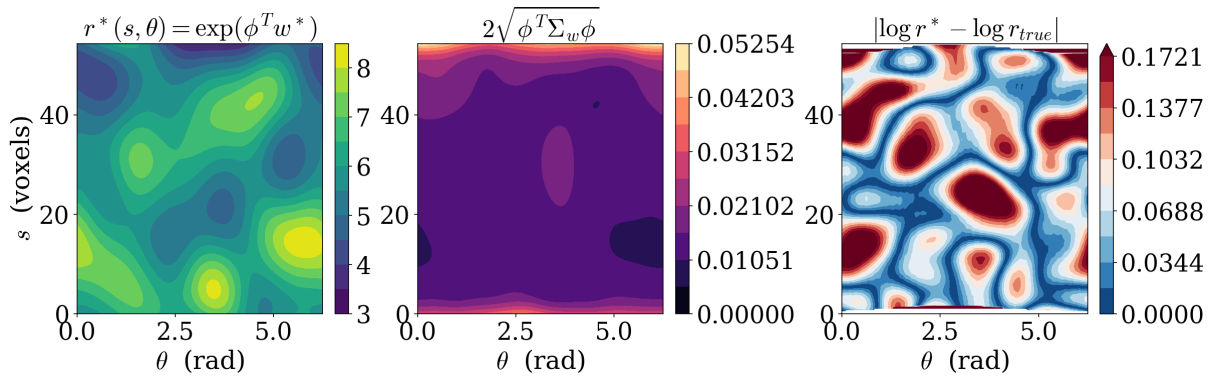


Figure 24: Uncertainty quantification for the jugular vein of evaluation case 1, over arc-length s (voxels) and azimuthal angle θ . Left: the MAP radius $r^*(s, \theta)$. Centre: the two-standard-deviation predictive band on the log-radius. Right: the absolute log-radius error $|\log r^* - \log r_{\text{true}}|$ against the ground-truth mask, on a scale reaching roughly three times the predicted band.

To demonstrate that the BETS algorithm applies to other tube shapes with noisy observations, we apply it to a scan of the aorta. This synthetic scan was produced by taking a binary aorta mask and adding correlated and uncorrelated Gaussian noise. Figure 25 shows the segmentation at two z values, along with 3D comparisons of the predicted and ground-truth segmentations.

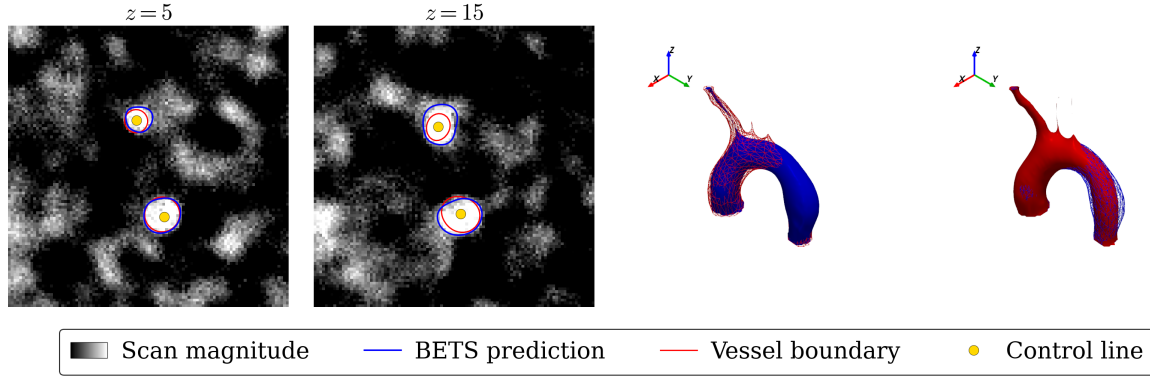


Figure 25: BETS applied to the synthetic aorta scan. Left two panels: axial cross-sections at $z = 5$ and $z = 15$ showing the scan magnitude. Right two panels: 3D renderings comparing the predicted surface (blue) with the ground-truth surface (red). One panel shows the vessel boundary as the wireframe, the other shows the predicted surface as the wireframe. The fit achieves Dice = 0.859.

The BETS prediction accurately tracks the ground-truth mask, but predicts an incorrectly wide right branch, and under-predicts the junctions at the top of the vessel.

On this scan the uncertainty strongly correlates with the error. Figure 26 plots the model’s uncertainty and error for the tube at the top of the aorta branch. BETS is highly uncertain about the incorrectly thin region. The two-standard-deviation error bounds are over 3 times too small though — again, this is due to the model mismatch: BETS under-estimates the probability of high-magnitude background regions because it wrongly assumes the noise is not correlated.

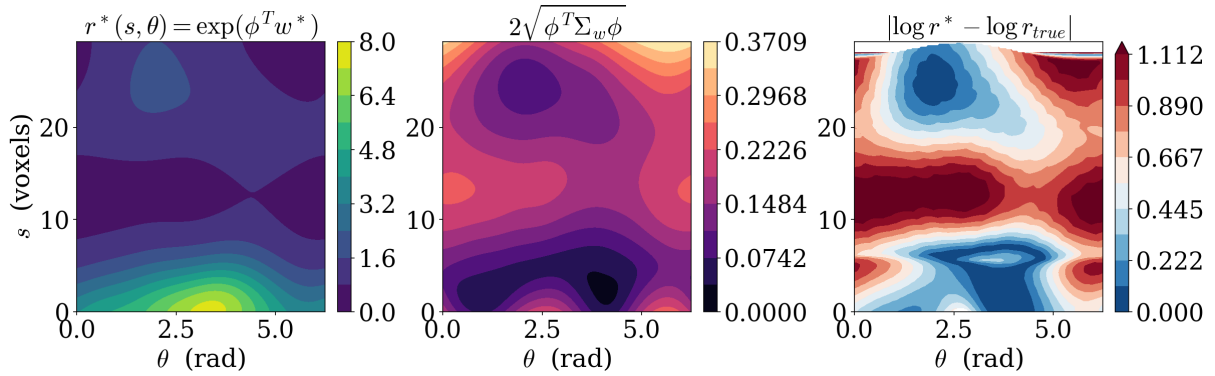


Figure 26: Uncertainty quantification for the tube at the top of the synthetic aorta, over arc-length s (voxels) and azimuthal angle θ . Left: the MAP radius $r^*(s, \theta)$. Centre: the two-standard-deviation predictive band on the log-radius. Right: the absolute log-radius error $|\log r^* - \log r_{\text{true}}|$ against the ground-truth mask. The high-uncertainty band (centre) coincides with the largest error (right).

3.5 Environmental, social and ethical impact

BETS is the initial-segmentation step in the blood vessel segmentation and blood flow reconstruction pipeline of [2], not a standalone tool. The impacts below follow from its use within that pipeline.

Benefits. Using BETS in a segmentation pipeline would bring environmental benefits since it means the same level of diagnostic information can be extracted from a shorter scan. This reduces the energy costs of imaging, which are significant — an MRI scanner consumes around 130 MWh per year [32].

BETS avoids ethical and regulatory issues around training data that are faced by models with a data prior, such as the U-Net. MRI scans are sensitive, and obtaining a patient's consent to include their flow MRI scan in a training dataset can be laborious. BETS does not use a training dataset.

BETS also brings social benefits. By making shorter scans feasible, it improves patient throughput and access to diagnostics.

Risks. To model unusual blood flows with rapid variation in the tube shapes, the GP hyperparameters must be set so this is feasible. There is a risk that an untrained doctor using the software may set these incorrectly.

There is potential for automation bias in the blood vessel segmentation pipeline, in which clinicians might over-trust the segmentations. The uncertainty bands are a partial mitigation of this bias. There is also a regulatory burden. Medicine is a safety-critical field, so any software needs extensive audit, validation and approval.

4 Conclusion

We demonstrated that while the BETS algorithm performs less well than the U-Net on the specific distribution that the U-Net was trained on, BETS is the only strategy to realistically segment the experimental scan — an out-of-distribution sample for the U-Net. We showed that foundation models for image segmentation are inappropriate for this problem.

The BETS algorithm was the main contribution of this project. The two most significant innovations we proposed were (i) applying a GP prior to the tube structure to make smooth tubes more probable, and (ii) using the Laplace EM algorithm instead of MCMC. Without these, the BETS approach would have failed. Using Laplace EM was arguably the most important design decision of the project because it enabled us to achieve approximate posterior inference without a sampling approach.

Although accurate, BETS is frequently over-confident because its likelihood assumes independent noise. Modelling the correlated background noise would solve this but make inference intractable unless significant approximations were applied.

The BETS algorithm has been designed to be coupled with the flow-reconstruction algorithm in [2]. This would produce an uncertainty-aware, efficient blood flow segmentation and reconstruction pipeline that acts on low-resolution noisy scans. This would allow doctors to extract diagnostic data from short scans, significantly increasing patient throughput for flow MRI scans.

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A U-Net hyperparameters

Hyperparameter	nnU-Net 3d_fullres	This work	Justification
Family	Plain U-Net	U-Net with residual blocks	The shortcut adds an identity path inside each stage, helping gradients flow at no parameter cost.
Encoder stages	5–6	4	Patch is 32^3 ; 4 stages keep the bottleneck at 4^3 , and a 5th stage would already shrink it to 2^3 .
Base channel width	32	32	
Max channels	320	256	One fewer stage caps the maximum at $32 \cdot 2^3 = 256$.
Downsample strides	(2, 2, 2) per stage	(2, 2, 2) per stage	
Normalisation	Instance norm	Instance norm	
Activation	LeakyReLU(0.01)	PReLU	This was the MONAI [20] default; the change is insignificant.
Deep supervision	Yes, $w_i = 1/2^i$	No	MONAI’s UNet does not expose intermediate logits.
Dropout	None	None	

Table 3: Architecture hyperparameters. The comparator is the 2021 nnU-Net 3d_fullres preset [12].

Hyperparameter	nnU-Net 3d_fullres	This work	Justification
Train / val pairs	5-fold CV (80%/20%)	720/80	Single 90%/10% split of the 800 synthetic pairs (-val_frac 0.1).
Optimiser	SGD, Nesterov, momentum 0.99	Adam [33]	The 1000-epoch SGD recipe is tuned for very long training; Adam converges faster at this scale with less hyperparameter sensitivity.
Initial learning rate	10^{-2}	10^{-3}	Adam-appropriate; nnU-Net's 10^{-2} is calibrated to SGD.
LR schedule	$(1 - e/1000)^{0.9}$ polynomial	Constant	Warming up or decaying inside 200 epochs has little effect.
Weight decay	3×10^{-5}	0	The generator already supplies regularisation through stochastic geometry.
Epochs	1000	200	720 small synthetic volumes; val Dice plateaus before epoch 40.
Iterations per epoch	250 (fixed)	180	We use the natural epoch (720/4); nnU-Net decouples epoch length from dataset size.
Batch size	Planner-derived (~ 2 at 128^3)	4	A 32^3 patch fits four samples per batch in VRAM.
Patch size	128^3 – 192^3 (planner-derived)	32^3	Evaluation volumes are 32–64 voxels per axis; larger patches would force whole-volume crops.
Foreground oversampling	33% of patches	None (uniform crop)	A 32^3 crop of a 32^3 – 64^3 volume has a high foreground hit rate by default.
Loss	Dice + CE (per-level weighted)	Dice + CE	Same composition; only the deep-supervision weighting is omitted.

Table 4: Training schedule and loss. Comparator as in Table 3.

Hyperparameter	nnU-Net 3d_fullres	This work	Justification
Resampling	To dataset-median spacing	None	Synthetic data is generated on the target grid; voxel spacing is 1 by construction.
Intensity normalisation	Per-case z-score	Per-case z-score	
Spatial: rotation	$\pm 30^\circ$ per axis	90° on the axial plane only	The generator already produces orientation variability.
Spatial: elastic	Random elastic	None	The generator morphs the SDF via random-field level-set evolution.
Spatial: scale	0.7–1.4	None	Built into the data generation pipeline.
Mirror	Random flip on all axes	Random flip on all axes	
Intensity: gamma	Yes	None	The generator already randomises contrast ($C \sim \mathcal{U}(3, 10)$).
Intensity: brightness / contrast	Yes	None	Same reason.
Intensity: Gaussian noise	Yes	None	The generator applies Rician noise at variable σ .
Halo half-width Δ	—	$\Delta \sim \mathcal{U}(0.1, 0.4)$	Synthetic-data generator parameter; sets the tanh transition width between vessel and background in Equation 37.

Table 5: Preprocessing and augmentation. Comparator as in Table 3.

Hyperparameter	nnU-Net 3d_fullres	This work	Justification
Mode	Sliding window, Gaussian-weighted	Sliding window, Gaussian-weighted	
Overlap	0.5	0.5	
Test-time augmentation	Mirror over 2^3 orientations	None	Single forward pass; $\sim 8\times$ faster at a $\sim 1\%$ Dice cost.
Ensembling	5-fold cross-validated soft-vote	None (single model)	A single fold is comparable to the SAM and geometry-driven baselines, which are also single-model.
Configuration choice	Empirical CV between 2D, 3d_fullres and 3d_cascade	3d_fullres only	Volumes are small enough that 2D and cascade have no advantage.

Table 6: Inference settings. Comparator as in Table 3.

B Risk assessment retrospective

How well did it reflect the hazards actually encountered?

In the risk assessment for this project we stated that we did not expect to encounter any high-risk hazards. This was because of the computational and theoretical nature of the work.

The hazards we encountered were from computer use. They included physical risks such as eye strain from screen use or wrist strain from typing. We took the following steps to mitigate these risks.

- Short breaks after extended periods of computational work to avoid strain.
- Appropriate positioning of the screen and keyboard to avoid strain.

The initial hazard assessment was accurate and encompassed all hazards experienced in this project.

How would we assess risk differently?

The risk assessment captured safety risks only. With hindsight, we should have extended our risk analysis to cover general project risks. Some of the risks that materialised are listed below.

- Hardware/software compatibility — the official nnU-Net implementation was not supported on the available AMD GPU.
- Compute access — insufficient compute prevented fine-tuning MedSAM 2 on flow MRI data.
- Data availability — no labelled real scans were obtained, which limited the real-world evaluation to qualitative comparison only.